Fully Automated Multi-Field Implementation of Elastoplasticity in the Framework of Generalized Standard Materials

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When you grow up you tend to get
told that the world is the way it is and
your life is just to live your life inside
the world. Try not to bash into the
walls too much. Try to have a nice
family, have fun, save a little money.
That's a very limited life. Life can be
much broader once you discover one
simple fact: everything around you
that you call life was made up by
people that were no smarter than you
and you can change it, you can
influence it... Once you learn that,
you'll never be the same again.

- Steve Jobs

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List of Symbols

Abbreviation	**Meaning**
PDE	Partial Differential Equation
FEA	Finite Elements Analysis
FEM	Finite Elements Method
DOF	Degrees of Freedom
GSM	Generalized Standard Materials
GP	Gauss Point
CPU	Central Processing Unit
GPU	Graphics Processing Unit
UFL	Unified Form Language
MPI	Message Passing Interface
JIT	Just In Time
FFCx	FEniCSx Form Compiler
PETSc	Portable, Extensible Toolkit for Scientific Computation
SNES	Scalable Nonlinear Equations Solver
MUMPS	MUltifrontal Massively Parallel Solver

Symbol	**Meaning**	**Unit**
$\boldsymbol{u}$	displacement field	$[m]$
E	Young's modulus	$[Pa]$
h	grid size in the square mesh	$[m]$
Δt	temporal discretization or time step	$[s]$
Ω	domain	$[m^3]$
$\boldsymbol{\varepsilon}$	strain tensor	$[m/m]$
$\boldsymbol{\sigma}$	stress tensor	$[Pa]$
$\boldsymbol{\alpha}_i$	set of internal state variables	[m/m]
$\boldsymbol{\mathcal{A}}_i$	set of driving forces	$[Pa]$
$\mathcal{D}$	mechanical dissipation	$[W]$
λ	plastic multiplier	[]
ψ	free energy density potential	$[J]$
π	dissipation potential	$[J]$
ν	Poisson's ratio	[]
λ	first Lamè parameter	$[Pa]$
μ	second Lamè parameter	$[Pa]$
H	isotropic hardening modulus	$[Pa]$
K	kinematic hardening modulus	$[Pa]$
$\boldsymbol{t}$	traction load	$[N]$

Abstract

We begin by introducing the theoretical background of Generalized Standard Materials (GSM), a framework that ensures thermodynamic consistency in material modeling. By deriving constitutive equations from energy potentials, GSM naturally satisfies the first and second laws of thermodynamics, making it a robust and physically sound approach for modeling a wide range of materials, including elastoplasticity. The energy functions encapsulate the energy stored within a material under deformation and serve as the foundation for deriving stress from the Helmholtz free energy as well as plastic strain evolution from the dissipative potential. To illustrate the application of this framework, we introduce specific plasticity models, such as the Von Mises plasticity model with linear isotropic and kinematic hardening, as case studies for our implementation. Using the multifield approach, instead of employing standard return mapping, we derive the weak form and the underlying equations governing plastic behavior. The implementation is carried out using FEniCSx, a powerful platform for solving partial differential equations (PDEs) with finite element method (FEM). Throughout this process, we enhance fully automation using specific FEniCSx libraries such as UFL, DOLFINx and Basix for modeling elastoplasticity efficiently. To investigate the model, we analyze the problem of a thick-walled tube under internal pressure. Finally, we highlight the broader impact of this methodology in automating the implementation of plasticity models within the Generalized Standard Materials (GSM) framework. By formulating the problem in terms of energy potentials, our approach enables seamless integration of new material models without requiring additional modifications to the underlying computational framework.

§ 1 Introduction

The development of computational tools for modeling elastoplastic behavior in materials has been a cornerstone of modern mechanics and materials science, enabling to predict and analyze the complex responses of materials under various loading conditions. Traditional approaches often require significant manual effort to implement and validate new constitutive models, which can be both time-consuming and error-prone.

The motivation for developing a fully automated implementation stems from the inefficiencies and limitations of traditional workflows for implementing existing elastoplastic models, which involves labor-intensive steps such as programming the governing equations based on the material's constitutive laws or deriving and coding numerical integration schemes and stress update algorithms. Each step introduces opportunities for errors and inefficiencies. Moreover, as the complexity of the model increases, so does the effort required for implementation.

By leveraging the principles of Generalized Standard Materials (GSM) theory, the proposed work achieves full automation in the implementation of elastoplasticity. The defining feature of this approach is its remarkable flexibility: users need only specify the two fundamental potentials (the free energy and the dissipation potentials) that characterize the material behavior. By specifying these two potentials, one can fully characterize a wide range of elastoplastic behaviors, from classical von Mises plasticity to more complex models involving nonlinear hardening. While GSM theory ensures thermodynamic consistency, the automation in this framework is achieved by leveraging FEniCSx to handle the derivation and implementation of equations directly from the potentials.

Once these potentials are defined, the framework automatically derives the constitutive equations, implements the numerical integration schemes and integrates the model into a finite element analysis (FEA) environment. This approach ensures consistency and reduces the time and effort required for implementation, eliminating the need for manual coding of constitutive laws or their derivatives as shown in Fig.1.

The final stage in this framework is the material model discovery process. Based on the numerical results obtained, the original material model is updated to improve its accuracy. This leads to new versions of the free energy and dissipation functions. These new functions replace the original ones and the entire process repeats. Over multiple iterations, the model becomes more refined, capturing complex material behaviors more accurately.

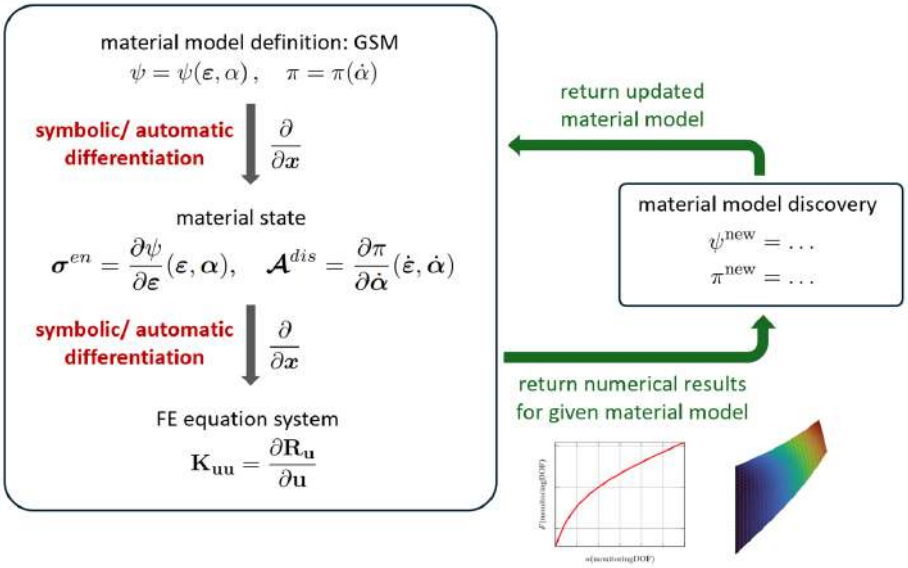

Figure 1: Computational framework for GSM models discovery. Image courtesy of Jonas Heinzmann

The work is structured as follows: in Section 3, we present the theory of plasticity, starting with the GSM framework, we discuss yielding criteria, particularly the von Mises yield criterion and describe the plastic flow rule. Finally, we introduce different hardening rules, including perfect plasticity, isotropic, kinematic, nonlinear and mixed hardening models. In Section 4, we detail the implementation, comparing the classical return mapping method with the multifield formulation. We explain the variational formulation of the multifield approach and its practical implementation using FEniCSx, covering aspects such as mesh generation (Gmsh), mixed elements, boundary conditions, class definitions, and the PETSc solver. In Section 5, we validate our approach with numerical examples, including perfect plasticity, linear isotropic hardening, linear kinematic hardening, linear mixed hardening and nonlinear hardening. These examples demonstrate the effectiveness and accuracy of the proposed methodology.

§ 2 State of the art

The classical and variational approaches to elastoplasticity offer distinct advantages and trade-offs. Classical computational plasticity methods rely on a two-level time integration strategy, with global integration for balance laws and local integration for constitutive equations. The integration process is referred to as Return Mapping algorithm and the local points at which it is performed are typically the quadrature points of a finite element. The local plastic state is then characterized by a set of internal variables associated with each quadrature point. This approach often requires additional Newton–Raphson procedures for complex yield functions, increasing computational overhead. In contrast, the variational framework derives balance and evolution equations from a unified principle, enabling single-level time integration and facilitating the use of higher-order time discretization methods. While this approach increases the number of unknowns, it eliminates the need for predictor-corrector methods and additional iterative schemes, making it a promising alternative for achieving higher accuracy and efficiency [42].

Furthermore, the multifield formulation naturally extends to gradient plasticity models, where additional degrees of freedom are introduced to regularize strain localization and capture size effects. In contrast, the return mapping algorithm cannot be directly applied to gradient plasticity, as it lacks the necessary framework to incorporate nonlocal terms. This makes the variational multifield approach a more flexible choice for advanced plasticity modeling.

By treating external and internal variables (e.g. plastic strain and its dual driving force [29]) as global unknowns, these formulations bypass the need for local integration at the quadrature point level, enabling efficient kernel formation for GPU acceleration [27]. GPU-accelerated material law evaluations further enhance computational efficiency, making the framework particularly suited for large-scale problems, with speedups of up to seven times compared to conventional CPU implementations [5].

Modern software frameworks like FEniCSx have been instrumental in simplifying the implementation of finite element solvers for solid mechanics problems involving complex constitutive models [Baratta et al. [2]]. These frameworks support multiple implementation strategies, including interpolation methods and external operators leveraging automatic differentiation with JAX [25].

External operators allow for the decoupling of constitutive models from the finite element framework, enabling the use of specialized libraries or external computational tools to evaluate material behavior. Instead of hardcoding the constitutive law directly within the finite element formulation, the solver calls an external function (or operator) that computes stresses, tangent moduli, or other relevant quantities.

However, these existing implementations do not go as far as our approach. Typically, external operators in finite element solvers focus on stress updates,

meaning that the best one can achieve is deriving the tangent operator from a stress update function or directly from the stress tensor. While this is a step toward automation, it still requires a predefined stress computation routine, which often relies on the classical return mapping formulation.

In contrast, our approach leverages the Generalized Standard Materials (GSM) framework, where constitutive models are defined directly in terms of their thermodynamic potentials (Helmholtz free energy and dissipation potential). By starting from these fundamental energy functions, rather than stress-strain relations, we achieve a higher level of automation: once the potentials are specified, all relevant quantities—including stress, plastic flow, and tangent moduli—are automatically derived. This eliminates the need for manually coding stress updates, making it a more elegant and generalizable approach for modeling plasticity and beyond.

Current approaches are specialized for specific tasks, such as computational efficiency or variational formulations, but lack a unified framework that integrates automation, non-local effect and multifield coupling under the principles of Generalized Standard Materials. The theoretical potential of GSM, particularly in automating the coupling of internal variables with thermodynamic potentials, remains under-explored in practical implementations.

§ 3 Theoretical Background

The scientific exploration of plasticity dates back to the mid-19th century, beginning with the work by [47], who proposed that a metal undergoes plastic yielding when the maximum shear stress reaches a critical threshold. [40] later applied Tresca's yield criterion to analyze stresses in partially plastic cylinders subjected to torsion or bending. Over the following decades, numerous experiments were conducted, leading to the proposal of various yield criteria. Among these, the most widely accepted was the one introduced by [48]. The Lévy-Mises stress-strain relations were experimentally validated by [28], who studied the deformation of metal tubes under combined tension and internal pressure. The theory was further refined by [37], who incorporated the elastic component of strain, building on an earlier idea proposed by [36]. Other significant contributions to the early development of plasticity theory include the works of [18], [31], [41], [19], and [10], among many others.

With the advent of powerful computers and the development of the finite element method between the 1970s and 1990s, contributions to plasticity theory expanded significantly. This progress highlighted the necessity of incorporating computational techniques to effectively solve the governing equations. Various numerical methods were introduced as extensions of the well-established elastic predictor-radial corrector algorithm, and a substantial body of research emerged to address these challenges [45], [9].

Nowadays, the theory of plasticity provides a broad framework for understanding the behavior of certain materials under stress. It focuses specifically on with solid materials that, when subjected to loading and subsequently unloading, may retain permanent (plastic) deformations. It is particularly relevant to scenarios where these enduring deformations are unaffected by the speed at which the loads are applied, a concept known as *rate-independent plasticity*.

Historically, the earliest plasticity models were phenomenological and were not formulated in accordance with the fundamental laws of thermodynamics. The incorporation of the Clausius-Duhem inequality, a direct consequence of the second law of thermodynamics, enabled the development of thermodynamically consistent plasticity models. To this end, [50] introduced an orthogonality principle as a generalization of Onsager reciprocal relations [34]. Building upon this foundation, [30] explored the hypothesis of a dissipation potential, which later led to an alternative formulation of plasticity theory. Following a different approach, [15] introduced the theory of Generalized Standard Materials. Unlike the classical formulation, which separately defines a yield function, a flow rule and a hardening law, this approach derives all three from a single potential. This class of materials is characterized by the fact that, once the yield function is specified, both flow and hardening laws are fully determined. In this work, we will address the theory of plasticity using the framework of Generalized Standard Materials, while also showing in detail how it satisfies each requirement of classical plasticity theory.

3.1 *Generalized Standard Materials*

Generalized Standard Materials[1] are defined by two convex[2], non-negative, scalar functions: the *Helmholtz free energy density potential* $\psi(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_i)$ which is continuously differentiable and zero at the origin (i.e. $\psi(\mathbf{0}, \mathbf{0}) = 0$) and the *dissipation potential* $\pi(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i)$ which is zero at the origin, but may not necessarily be continuously differentiable or strictly convex, meaning it can exhibit locally linear behavior. In this context, $\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \boldsymbol{u}^T + \nabla \boldsymbol{u})$ represent the small strain tensor while $\boldsymbol{\alpha}_i = \{\boldsymbol{\alpha}_I, ..., \boldsymbol{\alpha}_n\}$ denotes the set of internal state variables. These variables typically characterize the material's response to irreversible processes, such as plastic deformation or hardening. In the context of GSM, the specific choice of thermodynamic potentials, along with the appropriate parameter values, fully defines the material's behavior. Here, the notation and theoretical framework presented are consistent with those introduced in [12].

The convexity of the free energy potential plays a crucial role, as it ensures the ellipticity[3] of the governing partial differential equations (PDEs). These equations are solved to enforce the conservation of linear momentum, thereby establishing mechanical equilibrium. Additionally, the convexity of the flow potential guarantees that dissipation remains non-negative, thereby satisfying the second law of thermodynamics.

Consequently, the GSM framework offers significant advantages in material modeling because any material model adhering to this framework inherently complies with the principles of thermodynamics. As a direct consequence of these lasts combined with the mass and momentum balance equations from continuum mechanics, the Clausius-Duhem inequality, also known as the *dissipation inequality*, naturally arises (see Appendix A):

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \dot{\psi} \geq 0 \tag{1}$$

Using the chain rule, the time derivative of the Helmholtz potential $\dot{\psi}$ can be epressed as:

$$\dot{\psi} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} : \dot{\boldsymbol{\varepsilon}} + \frac{\partial \psi}{\partial \boldsymbol{\alpha}_i} : \dot{\boldsymbol{\alpha}}_i \tag{2}$$

Now, let's introduce the definitions of the energetic stress $\boldsymbol{\sigma}^{en}$ and the energetic driving forces $\boldsymbol{\mathcal{A}}_i^{en}$ as follows:

$$\boldsymbol{\sigma}^{en} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_i), \quad \boldsymbol{\mathcal{A}}_i^{en} = \frac{\partial \psi}{\partial \boldsymbol{\alpha}_i}(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_i) \tag{3}$$

These equations are known as the *state laws* because they define the thermodynamic forces associated with a given state of the system. By sequentially substituting equations (2) and (3) into (1), we derive the following expression:

$$\mathcal{D} = (\boldsymbol{\sigma} - \boldsymbol{\sigma}^{en}) : \dot{\boldsymbol{\varepsilon}} - \boldsymbol{\mathcal{A}}_i^{en} : \dot{\boldsymbol{\alpha}}_i = \boldsymbol{\sigma}^{dis} : \dot{\boldsymbol{\varepsilon}} + \boldsymbol{\mathcal{A}}_i^{dis} : \dot{\boldsymbol{\alpha}}_i \geq 0 \tag{4}$$

[1] The name refers to the generalization of the modeling framework for standard materials, i.e. for materials obeying the normality rule in plasticity (see Section 3.3).

[2] $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$, where $f(x, y) : \mathbb{R}^n \to \mathbb{R}$ and $\lambda \in [0, 1]$

[3] Let's consider a second-order linear PDE in two variables, written in the form $Au_{xx} + 2Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu + G = 0$ where A, B, C, D, E, F, and G are functions of x and y. The PDE is called elliptic if $B^2 - AC < 0$.

where the dissipative stress $\boldsymbol{\sigma}^{dis}$ and the dissipative driving forces $\boldsymbol{\mathcal{A}}_i^{dis}$ are defined as follows:

$$\boldsymbol{\sigma}^{dis} = \boldsymbol{\sigma} - \boldsymbol{\sigma}^{en}, \qquad \boldsymbol{\mathcal{A}}_i^{dis} = -\boldsymbol{\mathcal{A}}_i^{en} \tag{5}$$

State laws defined in (3) are not sufficient to completely describe the constitutive behavior. To achieve a complete description, they must be supplemented by the *complementary laws*

$$\boldsymbol{\sigma}^{dis} = \frac{\partial \pi}{\partial \dot{\boldsymbol{\varepsilon}}}(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i), \quad \boldsymbol{\mathcal{A}}_i^{dis} = \frac{\partial \pi}{\partial \dot{\boldsymbol{\alpha}}_i}(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i) \tag{6}$$

These equations describe how irreversible processes evolve in response to the associated thermodynamic forces. By substituting equations (6) into (4), we obtain:

$$\mathcal{D} = \frac{\partial \pi}{\partial \dot{\boldsymbol{\varepsilon}}} : \dot{\boldsymbol{\varepsilon}} + \frac{\partial \pi}{\partial \dot{\boldsymbol{\alpha}}_i} : \dot{\boldsymbol{\alpha}}_i \tag{7}$$

This clearly demonstrates that, regardless of the specific choice of π with the assumed properties, the dissipation $\mathcal{D}$ always remains non-negative, ensuring that the dissipation inequality is inherently satisfied.

The complete characterization of the material's response is achieved through the combined application of the state laws and complementary laws, which together define the *constitutive laws*:

$$\boldsymbol{\sigma} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_i) + \frac{\partial \pi}{\partial \dot{\boldsymbol{\varepsilon}}}(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i), \qquad \mathbf{0} = \frac{\partial \psi}{\partial \boldsymbol{\alpha}_i}(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_i) + \frac{\partial \pi}{\partial \dot{\boldsymbol{\alpha}}_i}(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i) \tag{8}$$

The second equation in (8) is commonly referred to as the *Biot equation* [3], named after Maurice Biot for his extensive contributions to the field. A detailed discussion of this equation can be found in Chapter 15 of [33].

3.1.1 Dual Dissipation Potential and Dual Complementary Laws

In some cases, it is more practical to define a dual dissipation potential that depends not on $(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}})$ but rather on the pair $(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis})$, as follows

$$\pi^\star = \pi^\star\left(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis}\right)$$

The dissipation potential and its dual are related through the Legendre-Fenchel transformation[4]:

$$\pi(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i) = \max_{\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis}} \left\{\boldsymbol{\sigma}^{dis} : \dot{\boldsymbol{\varepsilon}} + \boldsymbol{\mathcal{A}}_i^{dis} : \dot{\boldsymbol{\alpha}}_i - \pi^\star(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis})\right\} \tag{9}$$

$$\pi^\star\left(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis}\right) = \max_{\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i} \left\{\boldsymbol{\sigma}^{dis} : \dot{\boldsymbol{\varepsilon}} + \boldsymbol{\mathcal{A}}_i^{dis} : \dot{\boldsymbol{\alpha}}_i - \pi(\dot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\alpha}}_i)\right\} \tag{10}$$

While the stationarity conditions of equation (10) are given by (6), those of (9) read:

$$\dot{\boldsymbol{\varepsilon}} = \frac{\partial \pi^\star}{\partial \boldsymbol{\sigma}^{dis}}\left(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis}\right), \quad \dot{\boldsymbol{\alpha}}_i = \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_i^{dis}}\left(\boldsymbol{\sigma}^{dis}, \boldsymbol{\mathcal{A}}_i^{dis}\right) \tag{11}$$

[4]Let $f(x) : I \subset \mathbb{R} \to \mathbb{R}$ be a convex function. Then, the Legendre transform of f is the function $f(x)^\star : I^\star = \{x^\star \in \mathbb{R} : f^\star(x^\star) < \infty\} \subset \mathbb{R} \to \mathbb{R}$ defined by $f^\star(x^\star) = \sup_{x \in I}[x^\star x - f(x)]$

They describe the evolution of irreversible processes in relation to the corresponding thermodynamic forces. Essentially, these laws serve as evolution equations for both the strain and internal variables, often making them more practical than the equations in (6) (see Section 3.3 for further details).

Additionally, for solid materials, it is typically assumed that the dissipation potential is independent of the strain rate. Under this assumption, the Legendre-Fenchel transform takes the following form:

$$\pi^\star\left(\boldsymbol{\mathcal{A}}_i^{dis}\right) = \max_{\dot{\boldsymbol{\alpha}}_i}\left\{\boldsymbol{\mathcal{A}}_i^{dis} : \dot{\boldsymbol{\alpha}}_i - \pi(\dot{\boldsymbol{\alpha}}_i)\right\} \tag{12}$$

Hence, a simplified form of constitutive laws follows:

$$\boldsymbol{\sigma} = \frac{\partial\psi}{\partial\boldsymbol{\varepsilon}}(\boldsymbol{\varepsilon},\boldsymbol{\alpha}_i), \qquad \dot{\boldsymbol{\alpha}}_i = \frac{\partial\pi^\star}{\partial\boldsymbol{\mathcal{A}}_i^{dis}}\left(\boldsymbol{\mathcal{A}}_i^{dis}\right)\Bigg|_{\boldsymbol{\mathcal{A}}_i^{dis}=-\frac{\partial\psi}{\partial\boldsymbol{\alpha}_i}} \tag{13}$$

In this work, we will assume a dissipation potential that is independent of the strain rate. However, it is important to note that selecting a strain-rate-independent dissipation potential does not mean that the material model is entirely independent of the strain rate. Rate dependence is still preserved in the evolution of the internal variables. Within this framework, the dissipation inequality takes the following form:

$$\mathcal{D} = \boldsymbol{\mathcal{A}}_i^{dis} : \dot{\boldsymbol{\alpha}}_i = \frac{\partial\pi}{\partial\boldsymbol{\alpha}_i} : \dot{\boldsymbol{\alpha}}_i \geq 0 \tag{14}$$

3.2 *Yielding Criterion and Yield Surface*

The constrained evolution of stresses and thermodynamic forces is effectively captured by a convex scalar function f, known as the yield function. This function is negative when the material deforms elastically and becomes zero when plastic flow is about to begin. Therefore, plastic flow can only occur when

$$f\left(\boldsymbol{\mathcal{A}}_i^{dis}\right) = 0$$

where f is the scalar yield function which defines the *elastic domain* as the set

$$\mathscr{E} = \left\{\boldsymbol{\mathcal{A}}_I^{dis} \mid f\left(\boldsymbol{\mathcal{A}}_i^{dis}\right) < 0\right\}$$

of stresses for which plastic yielding is not possible. The convexity of this domain is ensured when f is a convex function of $\boldsymbol{\mathcal{A}}_i^{dis}$.

The set of stresses for which plastic yielding may occur constitutes the boundary of the elastic domain, represented by a hypersurface[5] within the stress space. This boundary is termed *yield surface*, and is defined as

$$\mathscr{P} = \left\{\boldsymbol{\mathcal{A}}_I^{dis} \mid f\left(\boldsymbol{\mathcal{A}}_i^{dis}\right) = 0\right\}$$

[5]High-dimensional generalization of a surface. Formally, in mathematics, a hypersurface in a n-dimensional space is a (n-1)-dimensional subset of that space.

3.2.1 Von Mises Yield Criterion

So far, the yield criterion has been addressed in its general form, without reference to any specific criterion. Now, we will focus on detailing the Von Mises yield criterion, which serves as the theoretical foundation for this work.

Many of the yield criteria proposed over the last two centuries are now considered of historical interest, as they conflict with more recent experimental observations that demonstrate the influence of hydrostatic stress on yielding. Among the simplest and most reliable criteria are those proposed by [47], who concluded that yielding occurs when the maximum shear stress exceeds a certain threshold and [48] in which is suggested that plastic yielding initiates when the second stress invariant J_2 (it is for this reason that plastic flow based on the von Mises yield criterion is often referred to as J_2 plasticity) reaches a critical value. The von Mises yield function is expressed as:

$$f = \sigma_{eq} - \sigma_y = \sqrt{3\,J_2} - \sigma_y = \sqrt{\frac{3}{2}\boldsymbol{\mathcal{A}}_{i,dev}^{dis} : \boldsymbol{\mathcal{A}}_{i,dev}^{dis}} - \sigma_y \tag{15}$$

where σ_{eq} and σ_y are the equivalent stress and the yield stress, respectively. We recall that

$$\boldsymbol{\mathcal{A}}_{i,dev}^{dis} = \boldsymbol{\mathcal{A}}_{i}^{dis} - \frac{1}{n}\mathrm{tr}(\boldsymbol{\mathcal{A}}_{i}^{dis})\mathbf{I}$$

represent the deviatoric dissipative driving forces, n the dimensionality of the problem and $\mathbf{I}$ is the second order identity tensor. Geometrically, equation (15) represents a cylinder in three-dimensional principal stress space with its axis aligned along the line where $\sigma_1 = \sigma_2 = \sigma_3$ (Fig. 2).

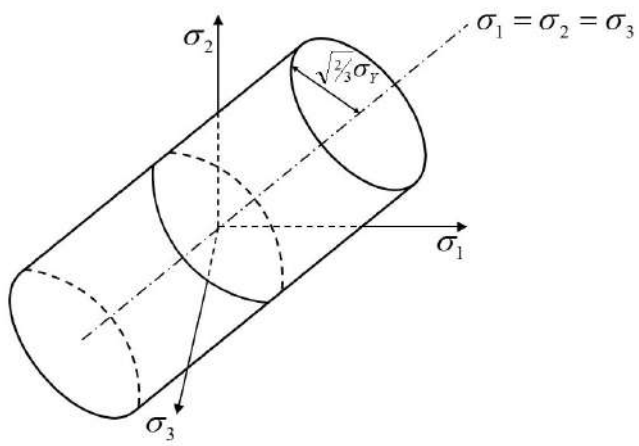

Figure 2: Von Mises yield locus, taken from Borja Erice and Dirk Mohr, Lecture Notes, Computational Modeling of Materials in Manufacturing

3.3 *Plastic Flow Rule*

Once the material reaches the yielding point, plastic flow initiates. For standard materials, the direction of this plastic flow is determined by the normality hypothesis, which states that the plastic strain rate is always perpendicular to the yield surface in stress space. This leads to an *associative plasticity model*, where the yield function and the dissipation potential coincide. As a result, the plastic flow rule reads:

$$\dot{\boldsymbol{\alpha}}_I = \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_I^{dis}} \left(\boldsymbol{\mathcal{A}}_I^{dis} \right) = \lambda \frac{\partial f}{\partial \boldsymbol{\mathcal{A}}_I^{dis}} \left(\boldsymbol{\mathcal{A}}_I^{dis} \right) \tag{16}$$

where λ is a non-negative scalar factor of proportionality, which is commonly known as *plastic multiplier*.

The direction of $\dot{\boldsymbol{\alpha}}_I$ is dictated by the gradient of $f(\boldsymbol{\mathcal{A}}_I^{dis})$, which points normal to the yield surface. This ensures that the dissipative driving force $\boldsymbol{\mathcal{A}}_I^{dis}$ contributes maximally to the dissipation energy $\mathcal{D}$, (defined in equation (14)), while satisfying $f(\boldsymbol{\mathcal{A}}_I^{dis}) = 0$. This alignment is a direct consequence of the associated flow rule and the principle of maximum dissipation. Therefore, for Generalized Standard Materials, the plastic flow rule corresponds to the dual complementary law in (11) for $i = I$.

Equation (16) assumes that the dissipation potential is continuously differentiable, which holds true for the von Mises criterion. However, many plasticity models are based on non-smooth dissipation potentials, such as the Tresca, Mohr–Coulomb, and Drucker–Prager models. In such cases, the dissipation potential is referred to as a *pseudo-potential* or *generalized potential* and the evolution laws can be handled by introducing the concept of subdifferential sets[6] (For an applied example, refer to Appendix B, or consult [21], [17], and [39]] for more details).

3.3.1 Prandtl-Reuss Equations

The Prandtl-Reuss law is the plasticity flow rule obtained by taking the von Mises yield function as dissipation potential:

$$\dot{\boldsymbol{\alpha}}_I = \lambda \frac{\partial f}{\partial \boldsymbol{\mathcal{A}}_I^{dis}} \left(\boldsymbol{\mathcal{A}}_I^{dis} \right) = \lambda \frac{\partial}{\partial \boldsymbol{\mathcal{A}}_I^{dis}} \left[\sqrt{\frac{3}{2} \boldsymbol{\mathcal{A}}_{I,dev}^{dis} : \boldsymbol{\mathcal{A}}_{I,dev}^{dis}} - \sigma_y \right]$$

$$\implies \qquad \dot{\boldsymbol{\alpha}}_I = \lambda \sqrt{\frac{3}{2}} \frac{\mathcal{A}_{I,dev}^{dis}}{\|\mathcal{A}_{I,dev}^{dis}\|} \tag{17}$$

The Prandtl–Reuss flow vector is a tensor parallel to the deviatoric projection of the dissipative driving forces tensor. This rule is typically applied alongside the von Mises yield criterion, forming what is known as the *von Mises associative model*.

[6]The *subdifferential* of a scalar function $y : \mathbb{R}^n \to \mathbb{R}$ at a point $\tilde{x}$ is the set:

$$\partial y(\tilde{x}) = \{ s \in \mathbb{R}^n \mid y(x) - y(\tilde{x}) \geq s \cdot (x - \tilde{x}), \quad \forall x \in \mathbb{R}^n \}$$

If the set ∂y is not empty at $\tilde{x}$, the function y is said to be *subdifferentiable* at $\tilde{x}$. The elements of ∂y are called *subgradients* of y.

3.4 *Hardening Rules*

As the material undergoes plastic deformation, its mechanical properties change, often manifesting as a shift in the geometry and position of the yield surface. The evolution of this initial yield surface during plastic deformation is controlled by the hardening rule. Various hardening models have been developed to accurately describe elastoplastic material behavior. Below, we briefly introduce some classical models, distinguishing between those with and without hardening.

3.4.1 Perfect Plasticity

A material model is considered perfectly plastic when it does not exhibits hardening, meaning that the yield stress σ_y remains constant and unaffected by the accumulated plastic deformation (Fig.3). In such cases, the yield surface, which represent the threshold beyond which plastic deformation begins, remains fixed. As the material deforms plastically, the yield surface does not expand or shift, indicating that the material's resistance to further deformation remains unchanged. It continues to undergo plastic deformation without any increase in the yield stress, meaning that no additional energy is required to sustain plastic flow beyond the initial yielding point.

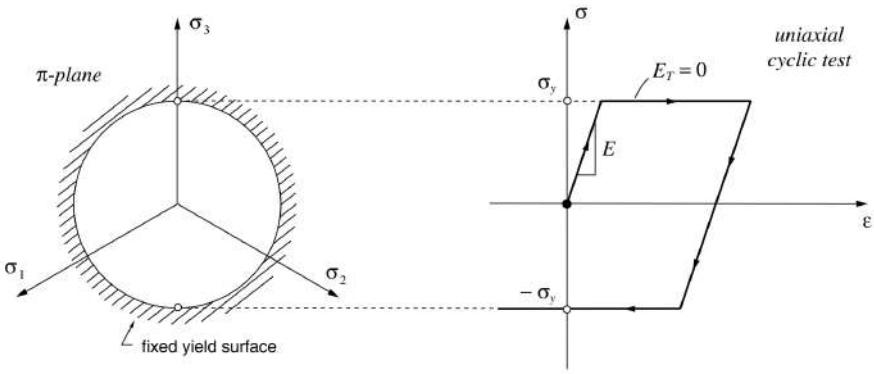

Figure 3: Uniaxial test and π-plane representation for a Von Mises perfect plasticity model, taken from De Souza Neto, Computational methods for plasticity (2008)

The two thermodynamic potentials that define the material behavior within the GSM framework for perfect plasticity are:

$$\psi(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_I) = \frac{1}{2}\lambda \operatorname{tr}(\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I)^2 - \mu(\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I) : (\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I) \tag{18}$$

$$\pi(\dot{\boldsymbol{\alpha}}_I) = \sigma_y \, \|\dot{\boldsymbol{\alpha}}_I\| \tag{19}$$

3.4.2 Isotropic Hardening

An isotropic hardening plasticity model is characterized by a uniform expansion of the yield surface in all directions during plastic deformation while maintaining

its shape, symmetry and orientation (Fig.4). This behavior signifies that the increase in yield stress occurs equally for all stress states, implying a uniform enhancement of the material's resistance to plastic deformation without any directional bias. The process is governed by the accumulation of equivalent plastic strain, resulting in a predictable and uniform increase in the material's capacity to resist yielding under further loading.

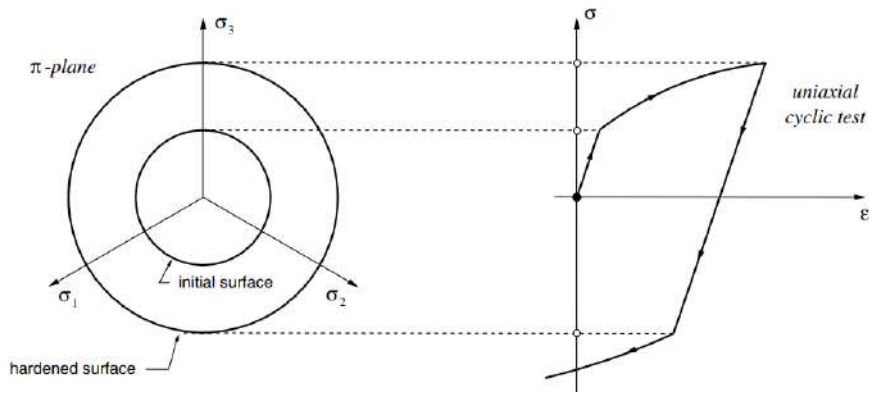

Figure 4: Uniaxial test and π-plane representation for a Von Mises plasticity model with isotropic hardening, taken from De Souza Neto, Computational methods for plasticity (2008)

Linear Isotropic Hardening

This model results in a proportional (linear) increase in the radius of the yield surface. In the case of von Mises loading surface (Section 3.2.1) the model is said to be linear isotropic hardening if:

$$\sigma_y(\boldsymbol{\alpha}_{II}) = \sigma_{y0} + H\boldsymbol{\alpha}_{II} = \sigma_{y0} - \mathcal{A}_{II}^{dis}$$

Given this evolution of the yield stress, the yield function can be expressed as the dual dissipation potential as:

$$\pi^\star\left(\boldsymbol{\mathcal{A}}_I^{dis}, \mathcal{A}_{II}^{dis}\right) = \sqrt{\frac{3}{2}\boldsymbol{\mathcal{A}}_{I,dev}^{dis} : \boldsymbol{\mathcal{A}}_{I,dev}^{dis}} + \mathcal{A}_{II}^{dis} - \sigma_{y0} \tag{20}$$

For this model, the two energy potentials are defined as follows:

$$\psi(\boldsymbol{\varepsilon}, \boldsymbol{\alpha}_I, \alpha_{II}) = \frac{1}{2}\lambda\,\mathrm{tr}(\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I)^2 - \mu(\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I) : (\boldsymbol{\varepsilon} - \boldsymbol{\alpha}_I) + \frac{1}{2}H\alpha_{II}^2 \tag{21}$$

$$\pi(\dot{\boldsymbol{\alpha}}_I, \dot{\alpha}_{II}) = (\sigma_y + H\alpha_{II})\|\dot{\boldsymbol{\alpha}}_I\| - H\alpha_{II}\dot{\alpha}_{II} \tag{22}$$

where H is the hardening modulus which dictates the rate of change of the yield stress with respect to the accumulated plastic strain α_{II} which is the scalar hardening variable capturing phenomena such as isotropic hardening effects. Note that perfect plasticity (defined in Section 3.4.1) is recovered by setting H=0.

3.4.3 Kinematic Hardening

While isotropic hardening effectively captures the expansion of the loading surface, it fails to describe phenomena that involve a rigid translation of the surface. A notable example is the Bauschinger effect, which cannot be modeled through simple isotropic hardening. In certain materials, an increase in yield stress in the direction of applied loading and plastic deformation leads to a reduction in yield stress in the opposite direction [26]. This phenomenon, known as the Bauschinger effect, reflects a form of directional anisotropy and plays a crucial role in accurately modeling cyclic loading behavior.

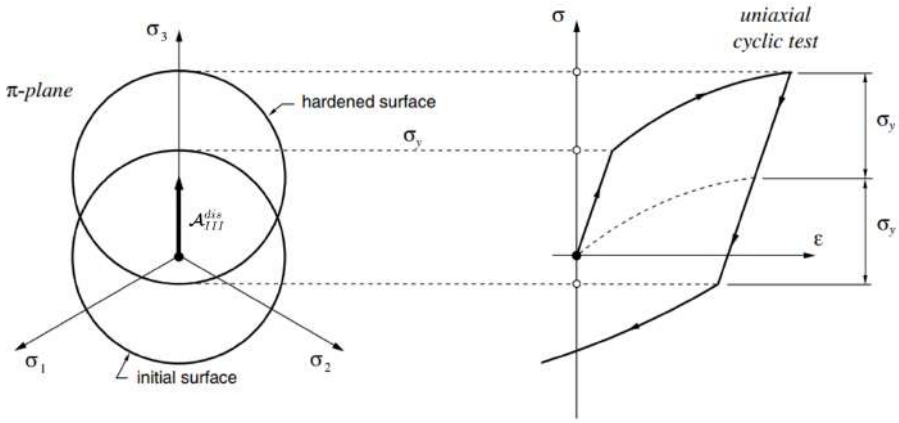

Figure 5: Uniaxial test and π-plane representation for a Von Mises plasticity model with kinematic hardening, modified based on De Souza Neto, Computational methods for plasticity (2008)

To model these type of hardenings, the kinematic hardening can be introduced as a rigid body translation of loading surface, while the rest of geometrical properties of loading surface remain unchanged (Fig.5. The loading surface of J2-plasticity with kinematic hardening as well as the dual dissipation potential can be described as follows:

$$\pi^{\star}\left(\boldsymbol{\mathcal{A}}_{I}^{dis},\boldsymbol{\mathcal{A}}_{III}^{dis}\right)=\sqrt{\frac{3}{2}\left(\boldsymbol{\mathcal{A}}_{I,dev}^{dis}+\boldsymbol{\mathcal{A}}_{III,dev}^{dis}\right):\left(\boldsymbol{\mathcal{A}}_{I,dev}^{dis}+\boldsymbol{\mathcal{A}}_{III,dev}^{dis}\right)}-\sigma_{y0} \quad (23)$$

where $\boldsymbol{\mathcal{A}}_{III}^{dis}$ is the thermodynamic force associated with kinematic hardening which governs the translation of the yield surface within the stress space.

Linear Kinematic Hardening

The thermodynamic potentials in the linear kinematic hardening model are defined as follows:

$$\psi(\boldsymbol{\varepsilon},\boldsymbol{\alpha}_I,\boldsymbol{\alpha}_{III})=\frac{1}{2}\lambda\,\mathrm{tr}(\boldsymbol{\varepsilon}-\boldsymbol{\alpha}_I)^2-\mu(\boldsymbol{\varepsilon}-\boldsymbol{\alpha}_I):(\boldsymbol{\varepsilon}-\boldsymbol{\alpha}_I)+\frac{1}{2}K\boldsymbol{\alpha}_{III}^2 \quad (24)$$

$$\pi(\dot{\boldsymbol{\alpha}}_I,\dot{\boldsymbol{\alpha}}_{III})=\sigma_y\|\dot{\boldsymbol{\alpha}}_I\|+K\boldsymbol{\alpha}_{III}(\dot{\boldsymbol{\alpha}}_I-\dot{\boldsymbol{\alpha}}_{III}) \quad (25)$$

where K is the kinematic hardening modulus, which controls the rate at which the yield surface translates in stress space.

§ 4 Implementational Approach

In the previous chapter, we reviewed the mathematical theory of plasticity and developed a general small-strain elastoplastic constitutive model within the framework of Generalized Standard Materials. Given the mathematical complexity of these constitutive theories, exact solutions to boundary value problems in practical engineering applications can only be obtained under highly simplified assumptions. Analytical solutions that are available are mostly limited to perfectly plastic models and are primarily used to determine limit loads and steady plastic flow for bodies with simple geometries [19]. To address these problems, this work employs the Finite Element Method for approximate solutions.

This chapter describes in detail the computational procedures necessary for the finite element solution of small strain plasticity problems within the framework of Chapter 3. To ensure broad applicability, the methodology presented is developed for elastoplasticity with any kind of hardening. The practical implementation of these procedures, including a thorough explanation of the algorithms and corresponding Python and C++ subroutines for FEniCSx, is demonstrated for the linear isotropic hardening model introduced in Section 3.4.2.

4.1 *Classical vs. Multifield Approach*

In the numerical modeling of material behavior, the strategies for integrating constitutive laws and solving the governing equations significantly influence the accuracy and efficiency of simulations. Two primary approaches for addressing these challenges are the classical method, which involves return mapping and local constitutive updates at the Gauss point level, and the multifield approach, where the field variables are solved all together.

The classical approach involves the decomposition of the numerical problem into two main levels: a global equilibrium problem that solves the balance of forces over the entire domain and a local constitutive problem solved at each integration point (Gauss point). The local constitutive problem is typically addressed using the return mapping algorithm, a numerical procedure that enforces consistency of the internal variables with the constitutive model. The main goal at the Gauss point level is to update the stress tensor and internal state variables based on the strain increments calculated from the global solution.

The development of this algorithmic scheme began with the classical radial return algorithm of [49], which was primarily applied to J2-flow theory. Notable extensions include the adaptation of the radial return method by [22] to incorporate linear isotropic and kinematic hardening, the midpoint return map proposed by [38] and the adaptation of these ideas to the plane stress problem by [44]. Additionally, alternative formulations of elastic-predictor/plastic-corrector methods can be found in [23]. Due to their significant industrial applications, these methods have garnered considerable attention in engineering literature

over the past decade. The current standard return mapping algorithms for classical plasticity are based on the backward Euler method (see Appendix C).

In contrast, the multifield approach involves the simultaneous solution of multiple field variables that are coupled in the governing equations. Instead of solving for displacements alone (as in standard finite element formulations), the multifield method may involve solving additional variables such as plastic strains or even higher-order quantities like damage or phase-fields. This simultaneous solution of multiple fields introduces additional equations into the system, allowing the global solver to account for more complex interactions between the mechanical fields directly. The coupling of these fields (in a highly nonlinear manner) can provide a more accurate representation of the physical processes, as the governing equations for each field are solved concurrently, thus capturing the mutual interactions between different phenomena. While these couplings can enhance the fidelity of simulations, they also increase the computational complexity. The simultaneous solution of multiple fields leads to a larger system of equations, which requires higher computational costs.

4.2 *Multifield Formulation*

The approach assumes that the balance of energy takes the form of a rate-type functional, whose stationarity condition yields the equations governing small-strain plasticity. Consequently, the material laws and the balance of linear momentum are solved at the same level at finite element nodes, utilizing continuous or discontinuous approximations. This results in a consistent mixed finite element strategy with a single time discretization scheme.

The primary advantage of adopting a multifield approach over a classical return mapping scheme is the enhanced flexibility and automation it offers through the definition of two GSM potentials. This methodology allows to fully automate the model implementation process, ensuring that when a new material model is defined, only the introduction of the new potentials is necessary. This significantly reduces the complexity involved in extending the model to accommodate new materials, while preserving consistency and accuracy across simulations.

In contrast to traditional return mapping approaches, which rely on stress-update algorithms and iterative correction schemes at quadrature points, the multifield framework derives the governing equations directly from thermodynamic potentials. This formulation ensures a unified treatment of balance laws and evolution equations, eliminating the need for separate local integration procedures. By embedding plastic evolution into a variational framework, the approach naturally extends to higher-order discretizations and gradient-enhanced plasticity models, which are challenging to handle with classical methods. Additionally, the multifield formulation avoids the complexity of manually enforcing consistency conditions for plasticity constraints, as they emerge naturally from the stationarity of the variational principle. This provides a more systematic and mathematically rigorous foundation for computational plasticity while maintaining flexibility for extending the model to more complex constitutive behaviors.

4.2.1 Variational Formulation

Variational principles rest on the idea that a physical system can be described by a specific functional and its stationary conditions achieved by differentiating this functional [16], [11], [24]. In plasticity, these principles are applied to varying extents. For instance, Ortiz proposed minimization principles to create update procedures for material laws [35]. Stationarity conditions are employed to generate weak forms at the structural level. When the functional involves multiple variables, distinct stationary conditions allow for the derivation of additional weak forms. This approach has been applied to plasticity and viscoplasticity in the work in [45] and [29]. In the following, we will extend a physically motivated variational principle to the elastoplastic case.

In the classical approach, elastoplastic material laws are incorporated implicitly by evaluating integrals at Gauss points. The standard solution procedure addresses both the structural and integral levels, each with its appropriate methods and time integration schemes. In this work, we adopted a variational procedure where displacements are just one component of the unknown field variables. Our approach builds on the work by [29] and [42].

To integrate the material model of Sect. 3 in the numerical context of a finite element analysis, the Neumann boundary conditions and the balance of linear momentum have to be formulated weakly. The starting point for this formulation is the principle introduced by Jourdain[20], which bridges the conceptual gap between D'Alembert principle of virtual work [8] and Gauss principle of least constraint [14]. Jourdain's principle asserts that the total virtual power $\Delta\mathcal{P}$ which encompasses contributions from the virtual internal power $\Delta\mathcal{E}$, the virtual power of inertial effects $\Delta\mathcal{K}$, the dissipative virtual power $\Delta\mathcal{D}$ and the virtual power due to external forces $\Delta\mathcal{W}$ is equal to zero.

$$\Delta\mathcal{P} = \Delta\mathcal{E} + \Delta\mathcal{K} + \Delta\mathcal{D} + \Delta\mathcal{W} = 0$$

In the context of this study, we restrict our analysis to quasi-static conditions. Under these conditions, the effects of inertia are negligible, allowing us to set $\Delta\mathcal{K} = 0$. Consequently, the total virtual power simplifies to include only the contributions from the remaining terms. Each of these terms is defined as follows:

$$\mathcal{E}(\boldsymbol{u}, \boldsymbol{\alpha}_i) = \int_{\mathcal{B}} \psi(\boldsymbol{\varepsilon}(\boldsymbol{u}), \boldsymbol{\alpha}_i)\,\mathrm{d}V \tag{26}$$

$$\mathcal{D}(\dot{\boldsymbol{\alpha}}_i) = \int_{\mathcal{B}} \pi(\dot{\boldsymbol{\alpha}}_i)\,\mathrm{d}V \tag{27}$$

$$\mathcal{W} = \int_{\partial\mathcal{B}} -\mathbf{t}\cdot\boldsymbol{u}\,\mathrm{d}A \tag{28}$$

We consider the evolution of the field variables over a finite time interval $[0, T]$, discretized into discrete time steps $t_0, t_1, \ldots, t_k$. The time increment between consecutive steps is denoted as $\Delta t = t_{n+1} - t_n$. The objective is to compute the constitutive state at the next time step t_{n+1}, given the known state at the current time step t_n. To achieve this, we use a single-step forward time discretization scheme.

The time derivatives of field quantities are approximated using a finite difference scheme:

$$\frac{d}{dt}(\cdot) = \dot{(\cdot)} \approx \frac{(\cdot) - (\cdot)^n}{\Delta t}$$

This simple linear-in-time evolution provides a basis for approximating the increments of stored energy and dissipated energy over the time step $[t_n, t_{n+1}]$:

$$\Delta\mathcal{E}(\boldsymbol{u}, \boldsymbol{\alpha}_i, \boldsymbol{u}^n, \boldsymbol{\alpha}_i^n) \approx \int_{t_n}^{t_{n+1}} \dot{\psi}(\boldsymbol{u}, \boldsymbol{\alpha}_i) \mathrm{d}\tau \approx \int_{\mathcal{B}} [\psi(\boldsymbol{u}, \boldsymbol{\alpha}_i) - \psi(\boldsymbol{u}^n, \boldsymbol{\alpha}_i^n)] \, \mathrm{d}V \tag{29}$$

$$\Delta\mathcal{D}(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_i^n) \approx \int_{t_n}^{t_{n+1}} \pi(\dot{\boldsymbol{\alpha}}_i) \mathrm{d}\tau \approx \int_{\mathcal{B}} \Delta t \, \pi \left(\frac{\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n}{\Delta t} \right) \mathrm{d}V \tag{30}$$

We now reformulate equation (30) using the dual dissipation potential, as defined in equation (12). This reformulation introduces an extended dissipation functional that incorporates an additional variable, the dissipative driving forces $\boldsymbol{\mathcal{A}}_i^{dis}$. The extended dissipation functional is expressed as:

$$\Delta\mathcal{D}^\star \left(\boldsymbol{\alpha}_i, \boldsymbol{\alpha}_i^n, \boldsymbol{\mathcal{A}}_i^{dis} \right) \approx \int_{\mathcal{B}} \left[\boldsymbol{\mathcal{A}}_i^{dis} : \left(\frac{\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n}{\Delta t} \right) - \pi^\star \left(\boldsymbol{\mathcal{A}}_i^{dis} \right) \right] \mathrm{d}V \tag{31}$$

The incremental work performed by external actions is given by:

$$\Delta\mathcal{W}(\boldsymbol{u}; \boldsymbol{u}^n) \approx \int_{t_n}^{t_{n+1}} \mathcal{W} \, \mathrm{d}\tau = \int_{\partial\mathcal{B}} - \boldsymbol{t} \cdot (\boldsymbol{u} - \boldsymbol{u}^n) \, \mathrm{d}A \tag{32}$$

In this formulation, we neglect contributions from non-homogeneous Neumann boundary conditions, body forces for the internal variables, and body forces for the displacement field. This simplification focuses the analysis on the essential contributions from external traction loads and displacement variations.

The canonical extended incremental variational principle for the rate-independent response now reads

$$\Delta\mathcal{P}^\star = \underbrace{\Delta\mathcal{E} + \Delta\mathcal{D}^\star}_{\Delta\mathcal{P}_{\mathrm{int}}} + \underbrace{\Delta\mathcal{W}}_{\Delta\mathcal{P}_{\mathrm{ext}}}$$

The variational principle seeks to find the stationary point of $\Delta\mathcal{P}^\star$ defined by the finite-step-sized incremental saddle point principle:

$$\{\boldsymbol{u}, \boldsymbol{\alpha}_i, \boldsymbol{\mathcal{A}}_i^{dis}\} = \arg \left\{ \inf_{\tilde{\boldsymbol{u}}} \inf_{\tilde{\boldsymbol{\alpha}}_i} \sup_{\tilde{\boldsymbol{\mathcal{A}}}_i^{dis}} \Delta\mathcal{P}^\star(\boldsymbol{u}, \tilde{\boldsymbol{\alpha}}, \tilde{\boldsymbol{\mathcal{A}}}_i^{dis}) \right\} \tag{33}$$

This principle implies that the first variation of $\Delta\mathcal{P}^\star$ must vanish:

$$\delta\Delta\mathcal{P}^\star = 0$$

which provides the weak form of the governing equations. To derive so, we compute the variation of the extended incremental potential $\Delta\mathcal{P}^\star$ with respect to each of the key variables: the displacement field $\boldsymbol{u}$, the internal variables $\boldsymbol{\alpha}$ and the dissipative driving forces $\boldsymbol{\mathcal{A}}^{dis}$. The first variation of $\Delta\mathcal{P}^\star$ with respect to

the displacement $\boldsymbol{u}$ captures the balance of linear momentum. Mathematically, this is expressed as:

$$\delta_{\boldsymbol{u}}\Delta\mathcal{P}^{\star} = \int_{\mathcal{B}} \left(\frac{\partial\psi}{\partial\nabla\boldsymbol{u}}\right) : \delta(\nabla\boldsymbol{u})\,\mathrm{d}V - \int_{\partial\mathcal{B}} \mathbf{t}\cdot\delta\boldsymbol{u}\,\mathrm{d}A$$

The first term in this expression represents the internal elastic forces, while the second term corresponds to the work done by external traction loads $\boldsymbol{t}$ on the boundary $\partial\mathcal{B}$. Substituting the definition of the elastic stress tensor from equation (3), we rewrite the variation as:

$$\delta_{\boldsymbol{u}}\Delta\mathcal{P}^{\star} = \int_{\mathcal{B}} \boldsymbol{\sigma}^{en} : \nabla(\delta\boldsymbol{u})\,\mathrm{d}V - \int_{\partial\mathcal{B}} \mathbf{t}\cdot\delta\boldsymbol{u}\,\mathrm{d}A \tag{34}$$

To simplify this expression, we apply the divergence theorem to the first term, which allows us to rewrite the integral over the volume $\mathcal{B}$ as a combination of surface and volume integrals:

$$\int_{\mathcal{B}} \boldsymbol{\sigma}^{en} : \nabla(\delta\boldsymbol{u})\,\mathrm{d}V = \int_{\partial\mathcal{B}} (\boldsymbol{\sigma}^{en}\cdot\boldsymbol{n})\cdot\delta\boldsymbol{u}\,\mathrm{d}V - \int_{B} \nabla\cdot\boldsymbol{\sigma}^{en}\cdot\delta\boldsymbol{u}\,\mathrm{d}V$$

Here, $\boldsymbol{n}$ denotes the outward unit normal vector on the boundary $\partial\mathcal{B}$. Substituting this result back into the variation, we find:

$$\delta_{\boldsymbol{u}}\Delta\mathcal{P}^{\star} = \int_{\partial\mathcal{B}} (\boldsymbol{\sigma}^{en}\cdot\boldsymbol{n} - \boldsymbol{t})\cdot\delta\boldsymbol{u}\,\mathrm{d}A - \int_{B} \nabla\cdot\boldsymbol{\sigma}^{en}\cdot\delta\boldsymbol{u}\,\mathrm{d}V \tag{35}$$

The first term vanishes if the boundary tractions satisfy the natural boundary condition $\boldsymbol{\sigma}^{en}\cdot\boldsymbol{n} = \boldsymbol{t}$. The second term leads to the strong form of the equilibrium equation:

$$\nabla\cdot\boldsymbol{\sigma}^{en} = 0 \quad \text{in}\,\mathcal{B}$$

Thus, the variation with respect to $\boldsymbol{u}$ enforces mechanical equilibrium within the body.

Next we consider the variation of $\Delta\mathcal{P}^{\star}$ with respect to the internal variables $\boldsymbol{\alpha}_i$. This variation governs the evolution of the internal state of the material. The expression for this variation is:

$$\delta_{\boldsymbol{\alpha}_i}\Delta\mathcal{P}^{\star} = \int_{\mathcal{B}} \left[\left(\frac{\partial\psi}{\partial\boldsymbol{\alpha}_i}\right) : \delta\boldsymbol{\alpha}_i + \boldsymbol{\mathcal{A}}_i^{dis} : \delta\boldsymbol{\alpha}_i\right]\mathrm{d}V$$

The first term represents the contribution of the elastic potential energy to the evolution of $\boldsymbol{\alpha}_i$, while the second term accounts for the dissipative forces. Substituting the definition of the energetic driving forces again from the equation (3), we rewrite the variation as:

$$\delta_{\boldsymbol{\alpha}_i}\Delta\mathcal{P}^{\star} = \int_{\mathcal{B}} \left[\boldsymbol{\mathcal{A}}_i^{en} : \delta\boldsymbol{\alpha}_i + \boldsymbol{\mathcal{A}}_i^{dis} : \delta\boldsymbol{\alpha}_i\right]\mathrm{d}V \tag{36}$$

By substituting the constitutive law (8), which relates $\boldsymbol{\mathcal{A}}_i^{en}$ and $\boldsymbol{\mathcal{A}}_i^{dis}$ we find:

$$\delta_{\boldsymbol{\alpha}_i}\Delta\mathcal{P}^{\star} = \int_{\mathcal{B}} \left(\boldsymbol{\mathcal{A}}_i^{en} + \boldsymbol{\mathcal{A}}_i^{dis}\right) : \delta\boldsymbol{\alpha}_i\,\mathrm{d}V = 0 \tag{37}$$

This indicates that the sum of the energetic and dissipative driving forces must vanish, providing the evolution equation for $\boldsymbol{\alpha}$ thus correctly reflecting the balance between the energetic and dissipative driving forces, aligning with the principles of thermodynamically consistent GSM formulation.

Finally, we compute the variation of $\Delta\mathcal{P}^\star$ with respect to the dissipative driving forces $\boldsymbol{\mathcal{A}}_i^{dis}$. This variation captures the dissipation mechanics and is given by:

$$\delta_{\boldsymbol{\mathcal{A}}_i^{dis}}\Delta\mathcal{P}^\star = \int_{\mathcal{B}} \left[(\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n) : \delta\boldsymbol{\mathcal{A}}_i^{dis} - \Delta t \, \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_i^{dis}} : \delta\boldsymbol{\mathcal{A}}_i^{dis} \right] \mathrm{d}V \tag{38}$$

Here, the first term represents the change in the internal variables between time steps, while the second term accounts for the dissipation potential $\pi^\star$. Factoring out the variation $\delta\boldsymbol{\mathcal{A}}_i^{dis}$ we obtain:

$$\delta_{\boldsymbol{\mathcal{A}}_i^{dis}}\Delta\mathcal{P}^\star = \int_{\mathcal{B}} \left[\left(\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n - \Delta t \, \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_i^{dis}} \right) : \delta\boldsymbol{\mathcal{A}}_i^{dis} \right] \mathrm{d}V \tag{39}$$

This implies that the dissipative forces must satisfy:

$$\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n - \Delta t \, \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_i^{dis}} = 0$$

By combining the variations with respect to $\boldsymbol{u}, \boldsymbol{\alpha}_i$ and $\boldsymbol{\mathcal{A}}_i^{dis}$ from equation (34), (37) and (39), the weak form of the governing equations is expressed as:

$$\begin{aligned} 0 \ni \delta\Delta\mathcal{P}^\star = \int_{\mathcal{B}} \Big[\boldsymbol{\sigma}^{en} : \nabla(\delta\boldsymbol{u}) + \left(\boldsymbol{\mathcal{A}}_i^{en} + \boldsymbol{\mathcal{A}}_i^{dis} \right) : \delta\boldsymbol{\alpha}_i \\ + \left(\boldsymbol{\alpha}_i - \boldsymbol{\alpha}_i^n - \Delta t \, \frac{\partial \pi^\star}{\partial \boldsymbol{\mathcal{A}}_i^{dis}} \right) : \delta\boldsymbol{\mathcal{A}}_i^{dis} \Big] \mathrm{d}V + \int_{\partial\mathcal{B}} \mathbf{t} \cdot \delta\boldsymbol{u} \, \mathrm{d}A \end{aligned} \tag{40}$$

This equation encapsulates the balance of momentum, the evolution of internal variables, and dissipation mechanics in their weak form. To solve these equations numerically, we discretize the weak form using finite element shape functions, leading to the formulation of residuals. Given the strong nonlinearity of these equations, we employ Newton's method, which involves linearization followed by an iterative solution process. The iterative update is governed by the following system of equations (omitting the iteration index for clarity):

$$\begin{bmatrix} \mathbf{K}_{\mathbf{uu}} & \mathbf{K}_{\mathbf{u}\boldsymbol{\alpha}_i} & \mathbf{K}_{\mathbf{u}\mathcal{A}_i^{dis}} \\ \mathbf{K}_{\boldsymbol{\alpha}_i\mathbf{u}} & \mathbf{K}_{\boldsymbol{\alpha}_i\boldsymbol{\alpha}_i} & \mathbf{K}_{\boldsymbol{\alpha}_i\mathcal{A}_i^{dis}} \\ \mathbf{K}_{\mathcal{A}_i^{dis}\mathbf{u}} & \mathbf{K}_{\mathcal{A}_i^{dis}\boldsymbol{\alpha}_i} & \mathbf{K}_{\mathcal{A}_i^{dis}\mathcal{A}_i^{dis}} \end{bmatrix} \begin{bmatrix} \Delta\mathbf{u} \\ \Delta\boldsymbol{\alpha}_i \\ \Delta\mathcal{A}_i^{dis} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{\mathbf{u}} \\ \mathbf{R}_{\boldsymbol{\alpha}_i} \\ \mathbf{R}_{\mathcal{A}_i^{dis}} \end{bmatrix}$$

This system corresponds to the equations that need to be defined in the software (FEniCSx), with the stiffness matrices and residuals forming the input for the solver. FEniCSx provides the tools to express these variational forms symbolically, define the finite element spaces for the state variables and automate the assembly of the linearized system. This numerical implementation solves for the updated state variables at each time step, adhering to the variational principles and boundary conditions established earlier.

4.3 *FEniCSx*

The FEniCS Project has long been recognized for its high-level automation of the FEM workflow, allowing users to specify PDEs in near-mathematical notation. The development of FEniCSx marks a significant shift when transitioning towards a modular architecture that decouples core components into separate libraries such as Basix for finite element basis function generation, UFL for variational formulation, DOLFINx as the computational back-end to assemble matrix and residual vector, FFcx for compilation of fast C code . The separation of components allows for more flexible and customized workflows. Users can now integrate external libraries for mesh generation (e.g. Gmsh) and linear algebra back-ends (e.g. PETSc) while maintaining compatibility with UFL and DOLFINx [1], [2].

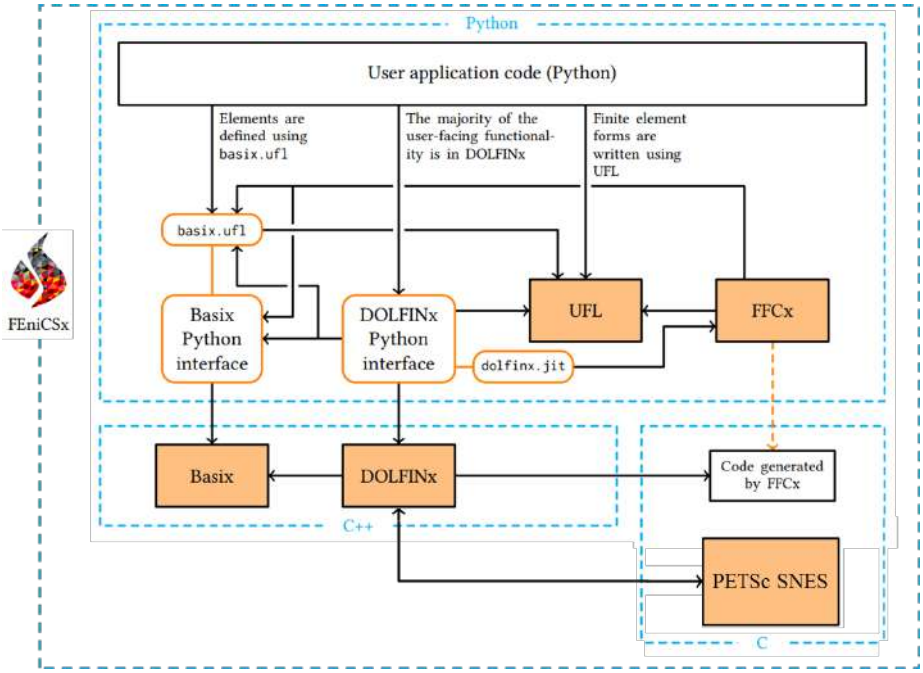

Figure 6: The interdependence of the core components of FEniCSx, modified based on Baratta, I. et al., DOLFINx: The next generation FEniCS problem solving environment (2023)

4.3.1 Mixed Elements

Basix is responsible for defining finite element basis functions. In FEM, basis functions are the building blocks used to approximate the solution of PDEs. Basix provides the tools for constructing these functions, including the mathematical definitions and their evaluation on the computational mesh. The Basix Python Interface allows users to interact with this functionality directly in Python. However, the computationally intensive tasks, such as evaluating basis functions and their derivatives, are implemented in C++ for efficiency. This sep-

aration ensures that users can define elements in a high-level language without sacrificing performance.

```
# create elements
vector_element = basix.ufl.element("CG", domain.basix_cell(), 1, shape=(2,))
tensor_element = basix.ufl.element("DG", domain.basix_cell(), 0, shape=(2, 2), symmetry=True)
scalar_element = basix.ufl.element("DG", domain.basix_cell(), 0, shape=())
mixed_element = basix.ufl.mixed_element(
    [vector_element,
     tensor_element, tensor_element,
     scalar_element, scalar_element,
     tensor_element, tensor_element]
)

# create function spaces
mixed_space = dolfinx.fem.functionspace(domain, mixed_element)
mixed_functions = dolfinx.fem.Function(mixed_space)
(u,
 α_I, A_I_dis,
 α_II, A_II_dis,
 α_III, A_III_dis
 ) = ufl.split(mixed_functions)
mixed_functions_old = dolfinx.fem.Function(mixed_space)
(u_old,
 α_I_old, A_I_dis_old,
 α_II_old, A_II_dis_old,
 α_III_old, A_III_dis_old
 ) = ufl.split(mixed_functions_old)

# test and trial functions
mixed_testfunctions = ufl.TestFunction(mixed_space)
(δu,
 δα_I, δA_I_dis,
 δα_II, δA_II_dis,
 δα_III, δA_III_dis) = ufl.split(mixed_testfunctions)
mixed_testfunctions_old = ufl.TestFunction(mixed_space)
(δu_old,
 δα_I_old, δA_I_old,
 δα_II_old, δA_II_dis_old,
 δα_III_old, δA_III_dis_old
 ) = ufl.split(mixed_testfunctions_old)

mixed_trialfunctions = ufl.TrialFunction(mixed_space)
```

Figure 7: Element Snipppet

As shown in Fig.7 we begin by creating finite element spaces for the problem variables:

- **Vector elements**: Created with `"CG"` (Continuous Galerkin) for vector fields with shape `(2,)`
- **Tensor elements**: Created with `"CG"` (Discontinuous Galerkin) for tensor fields with shape `(2,2)` and `symmetry = True` to reduce the total number of DOFs.
- **Scalar elements**: Also `"DG"` elements, used for scalar fields

These elements are combined into a mixed element using `basix.ufl.mixed_element`. Mixed elements are particularly useful for problems involving

multiple variables, as they allow the simultaneous representation of vector, scalar and tensor fields within a single finite element space. The mixed element is used to define a function space with `dolfinx.fem.FunctionSpace`. This space accommodates the problem's variables, which are represented as functions, i.e., state variables and mixed functions, both new and old. By splitting the mixed function into its components using `ufl.split`, the code separates the variables for easier manipulation in subsequent calculations.

Test and trial functions are defined next:

- Test functions (`mixed_testfunctions`) represent the virtual displacements or variations used in the weak form of the equations an will be used to integrate the residuals.
- Trial functions (`mixed_trialfunctions`) represent the increments or updates to the solution, so, the unknowns being solved for.

Mixed elements provide a powerful framework for simultaneously representing multiple fields within a single function space. This approach is especially advantageous for problems involving coupled systems, such as the one we are addressing, where external and internal dissipation variables are intrinsically linked through the governing equations. By using mixed elements, these variables are treated as part of a unified system, preventing errors that could arise from decoupling or approximating their interactions in staggered methods. Additionally, a monolithic formulation using mixed elements ensures that the Jacobian matrix of the system accounts for all couplings between variables, which enhances convergence and stability, particularly in the presence of strong nonlinearities.

However, selecting compatible elements for different fields is critical to avoid issues like locking. This constraint arises whenever the yield condition is pressure-insensitive such as in the case of the von Mises yield criterion. The issue of locking was first highlighted in [32] and has been extensively studied since. In particular, low-order Galerkin finite element methods can lead to an over-constrained pressure field, resulting in overly stiff or locked numerical solutions. To address this, techniques like classical LBB-satisfying displacement/pressure mixed finite element methods as reviewed in [6]), higher-order Galerkin finite element methods (such as the high-order triangle method of Scott [43]) and specialized three-field mixed finite element methods (like those in [46]) have been developed.

For mixed finite element methods, the choice of elements must satisfy the so-called inf-sup (or Ladyzhenskaya–Babuška–Brezzi, LBB) condition. It ensures the stability and accuracy of the numerical formulation, particularly in problems with constraints or coupled variables. The LBB condition provides a mathematical framework that balances the relationship between the finite element spaces used for primary and secondary variables. If this balance is not achieved, numerical instability can arise, manifesting as spurious oscillations or failure to converge to the correct physical solution.

The LBB condition mandates that the finite element space for the secondary variable must be sufficiently "large" relative to that of the primary variable. This

ensures that the constraints imposed by the governing equations can be accurately enforced without introducing numerical artifacts. In problems involving incompressible elasticity or plasticity, where displacement and internal variables are coupled, the choice of finite element spaces directly influences the stability and accuracy of the solution, as we will explore in the following chapter.

In addition, due to the dissipation potential in the multifield ansatz, a saddle point problem is formulated, as seen in equation (33). This formulation necessitates a thorough analysis of the existence and uniqueness of the solution, particularly in relation to the LBB condition, and it presents a specific challenge concerning solver application in the sense that solvers capable of handling saddle point problems typically employ specialized techniques such as preconditioning, iterative solvers and efficient matrix factorization methods to deal with the inherent complexity and ensure convergence.

4.3.2 Boundary Conditions

The next step of the implementation is to define boundary conditions. We did it through the DOLFINx Python Interface of FEniCSx.

```
# Get subspace of mixed space
u_sub_mixed_space = mixed_space.sub(0)
Q, _ = u_sub_mixed_space.collapse()


# left dofs and BCs
facets_left = dolfinx.mesh.locate_entities_boundary(
    domain, domain.topology.dim - 1, left
)
dofs_ux_left = fem.locate_dofs_topological(
    u_sub_mixed_space.sub(0), domain.topology.dim - 1, facets_left
)
ux_left = dolfinx.default_scalar_type(0.0)
bc_u0 = fem.dirichletbc(ux_left, dofs_ux_left, u_sub_mixed_space.sub(0))


# bottom dofs and BCs
facets_bottom = dolfinx.mesh.locate_entities_boundary(
    domain, domain.topology.dim - 1, bottom
)
dofs_uy_bottom = fem.locate_dofs_topological(
    u_sub_mixed_space.sub(1), domain.topology.dim - 1, facets_bottom
)
uy_bottom = dolfinx.default_scalar_type(0.0)
bc_u1 = fem.dirichletbc(uy_bottom, dofs_uy_bottom, u_sub_mixed_space.sub(1))

bcs_u = [bc_u0, bc_u1]
```

Figure 8: Boundary Conditions Snippet

The snippet in Fig.8 begins by using `dolfinx.mesh.locate_entities-`

`boundary()` to find the boundary facets. This function is part of the Dolfinx mesh module, which provides tools to handle geometric and topological aspects of the domain.

The Dolfinx FEM module (`dolfinx.fem`) is used to locate degrees of freedom associated with the boundaries. Specifically, `dolfinx.fem.locate_dofs_topological()` maps the boundary facets to the DoFs in the function space. This method ensures the boundary conditions are applied at the correct locations in the finite element space.

`mixed_space.sub()` is used to extract specific components of the mixed space:

- `mixed_space.sub(0)` corresponds to the u_x-displacement.
- `mixed_space.sub(1)` corresponds to the u_y-displacement.

The `collapse()` method, part of Dolfinx, simplifies the subspace into a standalone scalar or vector space. This is useful for further operations, like applying boundary conditions or solving sub-problems.

Finally, Dirichlet boundary conditions are imposed using `dolfinx.fem.dirichletbc()` and put into the list `bcs_u = [bc_u0, bc_u1]` which is a standard construct for managing multiple boundary conditions. This structure ensures that all conditions are applied consistently during the assembly process.

In the code snippet in Fig.9 a Neumann boundary condition (traction load) on the inner boundary of the domain is applied. The use of dolfinx and ufl ensures that the boundary is properly marked, integrated, and incorporated into the weak form.

The first part of the snippet in Fig.9 defines boundary markers. Each boundary is assigned an integer marker (1, 2, 3, 4) and the corresponding function (`lambda x: ...`) identifies points belonging to the boundary. Next, the code collects facet indices and assigns boundary markers:

- `mesh.locate_entities(domain, fdim, locator)` finds the mesh facets corresponding to the given boundary condition.
- `np.full_like(facets, marker)` assigns the correct boundary marker to each facet.

Finally, the facet markers are organized and stored and with `mesh.meshtags()` these markers are assigned to the mesh, allowing them to be referenced later.

Later on, integration measures needed for weak form assembly and a traction load are defined. In particular, the loading consists of a uniform pressure on the internal boundary. It will be progressively increased from 0 to a value slightly larger than

$$q_{lim} = \frac{2}{\sqrt{3}}\sigma_{y0} \log\left(\frac{R_e}{R_i}\right)$$

which is the analytical collapse load for a perfectly-plastic material [4]. Finally, the Neumann boundary condition is applied through the definition of `traction_inner(δu_)`:

- `q_loading=fem.Constant(domain, 0.0 * q_lim)` defines a constant traction load in the weak form.
- `ds(2)` indicates that this integral is evaluated over the boundary marked with "2", corresponding to the inner circular boundary.

```
# create edge markers
boundaries = [
    (1, lambda x: np.isclose(x[0], 0)),
    (2, lambda x: np.isclose(x[0] ** 2 + x[1] ** 2, Ri**2),),
    (3, lambda x: np.isclose(x[0] ** 2 + x[1] ** 2, Re**2),),
    (4, lambda x: np.isclose(x[1], 0)),
]

facet_indices, facet_markers = [], []

for marker, locator in boundaries:
    facets = mesh.locate_entities(domain, fdim, locator)
    facet_indices.append(facets)
    facet_markers.append(np.full_like(facets, marker))

facet_indices = np.hstack(facet_indices).astype(np.int32)
facet_markers = np.hstack(facet_markers).astype(np.int32)
sorted_facets = np.argsort(facet_indices)
facet_tags = mesh.meshtags(
    domain, fdim, facet_indices[sorted_facets], facet_markers[sorted_facets]
)
domain.topology.create_connectivity(fdim, ndim)


# normals and measures for integration
n = ufl.FacetNormal(domain)
dx = ufl.Measure("dx", domain)
ds = ufl.Measure("ds", domain, subdomain_data=facet_tags)


# traction load
q_lim = (2.0 / np.sqrt(3)) * np.log(Re / Ri) * GSM.σ_y
q_loading = fem.Constant(domain, 0.0 * q_lim)

def traction_inner(δu_):
    return -q_loading * ufl.inner(n, δu_) * ds(2)
```

Figure 9: Loads Snippet

4.3.3 Class Definition

This is a crucial aspect of the numerical implementation. By defining a class for each type of model (Fig.10, we ensure that adding a new model is straightfor-

ward, only an additional class needs to be created. This class would define the new thermodynamic potentials that describe the material behavior and specify the material parameters to be used. Other sections of the code, which were previously established, remain unchanged, ensuring that the implementation is fully automated and scalable.

```
class LinearHardening:

    def __init__(self):

        # Elastic parameters
        self.E = 70e3
        self.ν = 0.3
        self.λ = self.E * self.ν / (1 + self.ν) / (1 - 2 * self.ν)
        self.μ = self.E / 2.0 / (1 + self.ν)

        # Plastic parameters
        self.σ_y = 250.0
        self.E_tan = self.E / 100.0
        self.H = self.E * self.E_tan / (self.E - self.E_tan)
        self.K = 2000


    # Free energy potential
    def ψ(self, ε_, α_I_, α_II_, α_III_):
        return (
            0.5 * self.λ * ufl.tr(ε_ - α_I_) ** 2
            + self.μ * ufl.inner(ε_ - α_I_, ε_ - α_I_)
            + 0.5 * self.H * α_II_**2
            + 0.5 * self.K * ufl.inner(α_III_, α_III_)
        )


    # Dual dissipation rate potential
    def π_star(self, A_I_dis_, A_II_dis_, A_III_dis_, ndim):
        A_I_dis_dev = A_I_dis_ - 1 / ndim * ufl.tr(A_I_dis_) * ufl.Identity(ndim)
        A_III_dis_dev = A_III_dis_ - 1 / ndim * ufl.tr(A_III_dis_) * ufl.Identity(ndim)
        A_dis_norm = ufl.sqrt(
            3 / 2 * (ufl.inner(A_I_dis_dev + A_III_dis_dev, A_I_dis_dev + A_III_dis_dev)))
        return ufl.conditional(
            A_dis_norm > self.σ_y - A_II_dis_, 1 / 2 * (A_dis_norm - self.σ_y + A_II_dis_) ** 2 , 0.0
        )
```

Figure 10: Class Snippet

4.3.4 PETSc Solver

Once the variational form is defined using UFL (Fig.11), it must be compiled into low-level code for efficient evaluation. This is the role of FFCx (FEniCSx Form Compiler). FFCx takes the forms written in UFL and generates optimized C code. This process, known as Just-In-Time (JIT) compilation, ensures that the code is tailored to the specific problem being solved. The integration with `dolfinx.jit` enables DOLFINx to dynamically compile and execute this code during runtime. This approach combines the flexibility of Python with the performance of compiled code.

```
# Weak form
F = (
    ufl.inner(σ_en(ε(u), α_I, α_II, α_III), ufl.grad(δu)) * dx
    + ufl.inner(A_I_en(ε(u), α_I, α_II, α_III), δα_I) * dx
    + ufl.inner(A_I_dis, δα_I) * dx
    + ufl.inner(A_II_en(ε(u), α_I, α_II, α_III), δα_II) * dx
    + ufl.inner(A_II_dis, δα_II) * dx
    + ufl.inner(A_III_en(ε(u), α_I, α_II, α_III), δα_III) * dx
    + ufl.inner(A_III_dis, δα_III) * dx
    + ufl.inner(α_I, δA_I_dis) * dx
    - ufl.inner(α_I_old, δA_I_dis) * dx
    + ufl.inner(α_II, δA_II_dis) * dx
    - ufl.inner(α_II_old, δA_II_dis) * dx
    + ufl.inner(α_III, δA_III_dis) * dx
    - ufl.inner(α_III_old, δA_III_dis) * dx
    - ufl.inner(dπ_star_dA_I_dis(A_I_dis, A_II_dis, A_III_dis), δA_I_dis) * Δt * dx
    - ufl.inner(dπ_star_dA_II_dis(A_I_dis, A_II_dis, A_III_dis), δA_II_dis) * Δt * dx
    - ufl.inner(dπ_star_dA_III_dis(A_I_dis, A_II_dis, A_III_dis), δA_III_dis) * Δt * dx
    - traction_inner(δu)
)


# Jacobian
J = ufl.derivative(F, mixed_functions, mixed_trialfunctions)
```

Figure 11: Weak Form Snippet

At the lowest layer of the diagram of Fig.6, we see the computational backend, which includes Basix, DOLFINx and PETSc SNES. These components handle the most computationally demanding tasks, such as matrix assembly, linear algebra operations, and the solution of nonlinear systems. The backend is implemented in C++ and C, ensuring that the framework can handle large-scale simulations efficiently.

In Fig. 12, a PETSc solver employing Newton's method with LU factorization (MUMPS) is presented. The LU factorization (`pc_type="lu"`) ensures a direct solution at each Newton step, maintaining strict tolerance criteria that guarantee precise convergence without requiring additional step control. In our case, LU decomposition was chosen because we are using mixed function spaces—different discretizations for different fields—which can result in a non-symmetric system matrix. Since symmetry is a necessary condition for Cholesky decomposition to function correctly, LU factorization was preferred as a more general and reliable alternative.

If convergence issues arise in more complex scenarios (e.g., highly nonlinear material models) a linesearch could be used, such as `snes_linesearch_type ="bt"` (backtracking) to adjust step sizes dynamically. Moreover, for larger problems, iterative solvers (`ksp_type="gmres" or "cg"`) with preconditioners (`pc_type = "gamg"`) can be adopted.

```
problem = SNES_solver.SNESProblem(F, J, mixed_functions, bcs_u)
X_vec = mixed_functions.x.petsc_vec.copy()
F_vec = dolfinx.fem.petsc.create_vector(problem._F)
J_mat = dolfinx.fem.petsc.create_matrix(problem._J)


# Solve
opts = PETSc.Options()
snes = petsc4py.PETSc.SNES().create(domain.comm)
snes.setFunction(problem.F, F_vec)
snes.setJacobian(problem.J, J=J_mat, P=None)
opts["snes_type"] = "newtonls"
opts["pc_type"] = "lu"
opts["pc_factor_mat_solver_type"] = "mumps"
opts["ksp_monitor"] = ""
opts["ksp_type"] = "preonly"
opts["snes_stol"] = 1.0e-7
opts["snes_rtol"] = 1.0e-7
opts["snes_atol"] = 1.0e-7
opts["snes_divergence_tolerance"] = 1e9
opts["snes_max_it"] = 200
opts["snes_max_fail"] = 200
opts["snes_monitor"] = ""
opts["snes_converged_reason"] = ""
snes.setFromOptions()
snes.setErrorIfNotConverged(True)
snes.logConvergenceHistory(norm=2)
```

Figure 12: Solver Snippet

§ 5 Numerical Examples

For the numerical analysis of the multifield implementation of small-strain elasto-plasticity, described in Chapter 4, various examples are considered. These serve both to verify the applicability of this variational approach and to compare its performance with the classical return mapping method in terms of solution accuracy and convergence properties. The geometry used in these examples is a thick-walled tube under internal pressure, as described in [4] and shown in Fig.13.

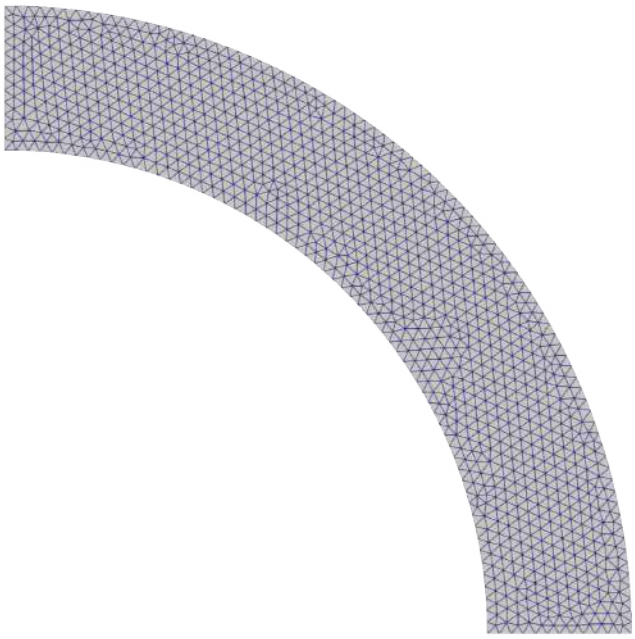

Figure 13: Mesh with triangular elements

5.1 *Perfect Plasticity*

As highlighted in the introduction of the previous chapter, analytical solutions for plasticity problems are relatively scarce. One of the few available solutions, which we use to validate our numerical approach, is well documented in [9]. Similar to the study in [4], a thick-walled tube is analyzed under plane strain conditions. However, since our work focuses on purely two-dimensional models, slight modifications have been made to the constitutive equations both in the analytical and the return mapping implementations.

The blue and green markers in Fig.14, representing the multifield and return mapping approaches, closely follow the analytical solution, confirming the accuracy of the numerical methods. Minor discrepancies may arise due to numerical

discretization errors; however, the overall agreement validates the computational models employed. This comparison primarily serves to validate the multifield approach. In the subsequent examples, we will analyze computational efficiency and convergence properties.

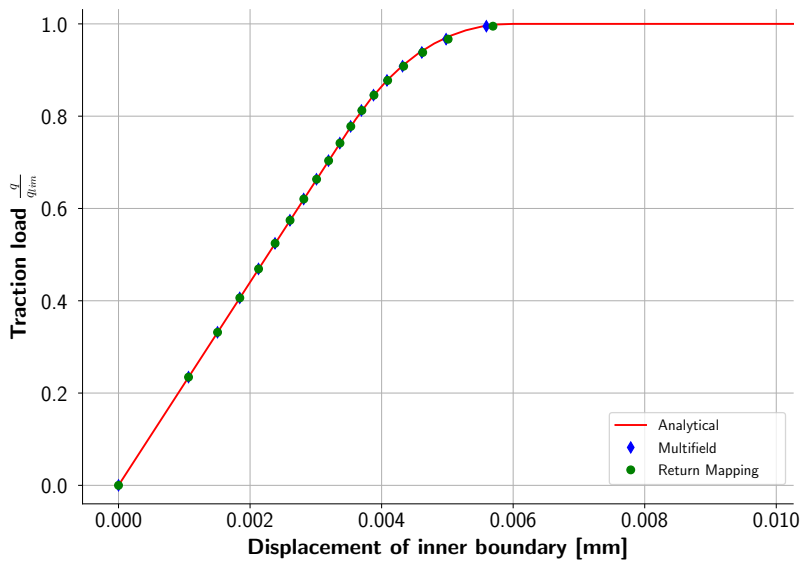

Figure 14: Comparison with analytical solution

5.2 *Linear Isotropic Hardening*

For this numerical example, the model described in Section 3.4.2 has been used (see Fig.10 to check the material parameters and the thermodynamic potentials which describe the material behavior), while a numerical comparison has been made with the return mapping implementation in [4].

A stable and efficient choice of elements involves using linear Continuous Galerkin (CG) elements for displacement and constant Discontinuous Galerkin (DG) elements for internal variables. This combination is particularly advantageous because it strikes a balance between stability and computational efficiency. Linear CG elements provide a smooth and continuous representation of displacement fields, while constant DG elements capture localized features in the internal variables without introducing excessive degrees of freedom.

The stability of this combination stems from the relative sizes of the finite element spaces. The DG space for internal variables, being "smaller" than the CG space for displacement, prevents over-constraining the system. In some cases, however, the problem at hand may require a more refined approach. In the following, a comparison between various degrees in the element choice is

performed:

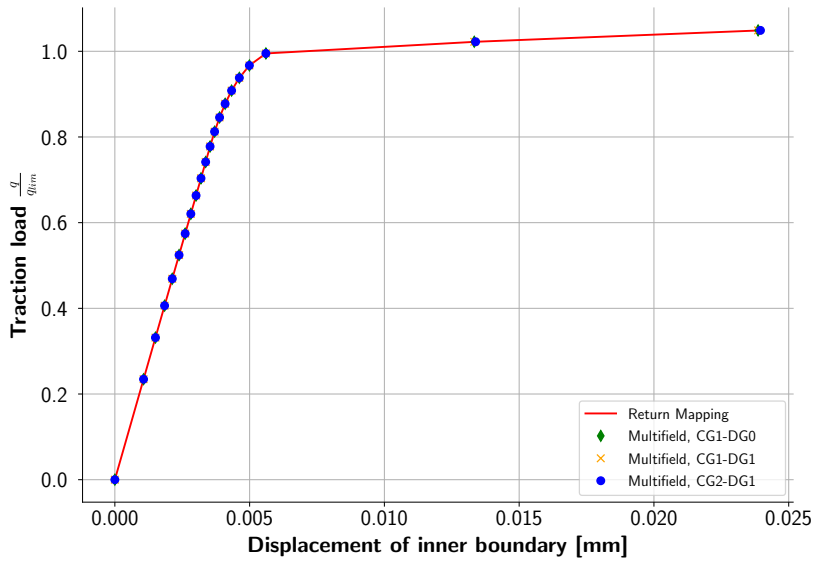

Figure 15: Element types: Return Mapping vs. Multifield

From Fig.15 is visible that in the linear elastic regime, all methods exhibit nearly identical results. This indicates that for linear problems, the choice of finite element pairings (e.g., CG1-DG0 vs. CG2-DG1) or numerical methods (Return Mapping vs. Multifield) does not significantly affect the solution. In the nonlinear behavior CG1-DG0 is more computationally efficient (Fig.16) while maintaining accuracy comparable to CG2-DG1, which is potentially more accurate, but it struggles with convergence during this phase, requiring higher computational effort that might be justified only if it provides additional benefits not visible in the force-displacement plot of Fig:15.

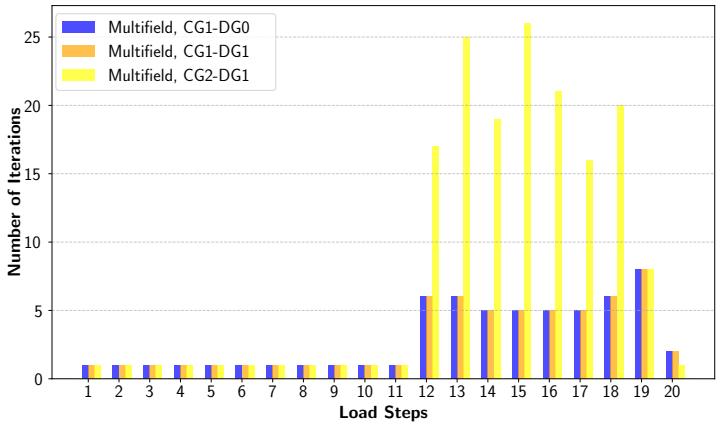

Figure 16: Number of iterations among different element types

The higher iteration count for CG2-DG1 visible in Fig. 16 suggests that its higher-order formulation does not provide a computational advantage for these relatively simple model. Additionally, when using the return mapping approach, there are 2 degrees of freedom (DOFs) per node, meaning that with P2 elements, each element has 12 DOFs (since a triangular P2 element has 6 nodes). In contrast, the multifield approach, which solves not only for displacement but also for internal variables, results in 10 DOFs per node (2 for $\boldsymbol{u}$, 3 for $\boldsymbol{\mathcal{A}}_I^{dis}$, 3 for $\boldsymbol{\alpha}_I$, 1 for $\mathcal{A}_{II}^{dis}$ and 1 for α_{II} leading to 60 DOFs per element. This number can be fairly reduced just by introducing a function space for symmetric tensors in the code as shown in Fig.7. Doing so, convergence is slightly better handled as visible from Fig.17.

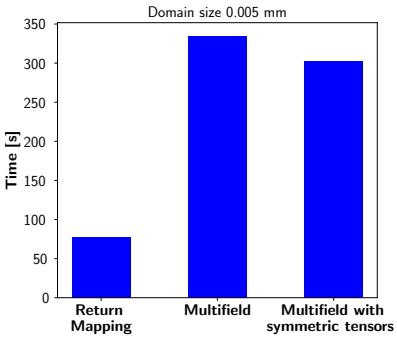

Figure 17: Runtime: Return Mapping vs. Multifield

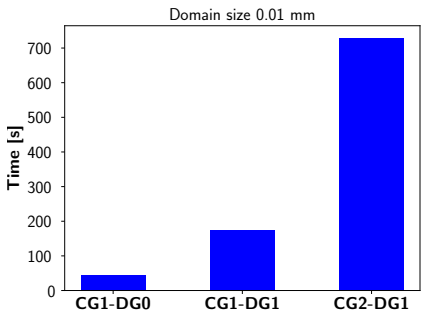

Figure 18: Runtime in Multifield: Different element types comparison

From the results shown in Fig.15 in and in Fig.18, it can be concluded that the optimal choice for this model is to use P1P0 elements, as this elements configuration provides a good balance between computational efficiency and accuracy relative to the reference solution.

The significant increase in computational effort found in Fig.18 becomes problematic when the mesh is coarsened, as it can lead to a violation of the inf-sup condition and cause instabilities when using P2P1 mixed elements. In well-resolved simulations, these elements generally provide a good balance: the higher-order P2 elements accurately capture deformation, while the P1 elements offer a compatible, albeit lower-order, approximation for pressure. However, on a coarse mesh, while the P2 elements may still approximate the displacement field well, the P1 pressure field struggles to capture the necessary variations. This imbalance can lead to deviations from the expected behavior or even divergence in the worst case. This suggests that the coarse mesh does not allow the P2P1 elements to fully satisfy the LBB condition, leading to instability and locking-like behavior. Let's see how changing the domain size the result changes:

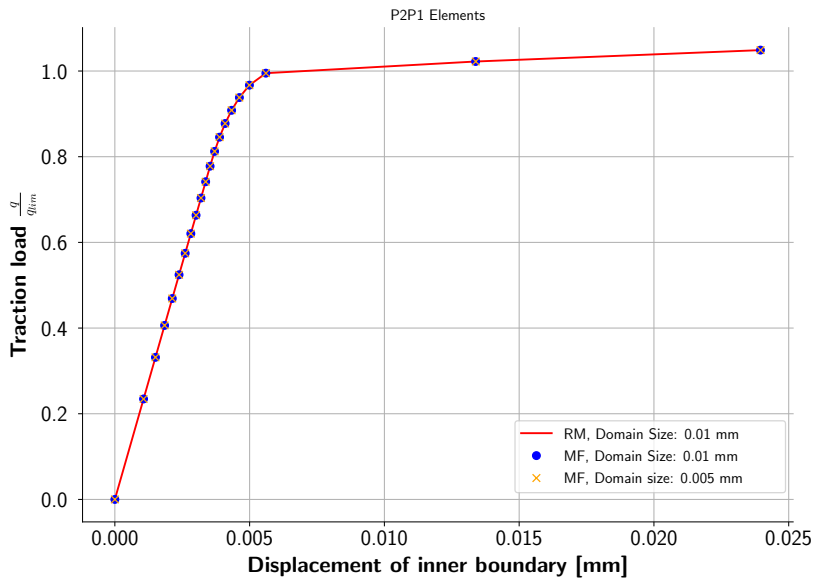

Figure 19: Domain sizes: Return Mapping vs. Multifield

Notably, refining the mesh does not always lead to improved results when using the same element type (P2P1). As shown in Fig.19, for mesh sizes finer than 0.01 mm, the results remain unchanged. Therefore, to minimize computational cost without sacrificing accuracy, a domain size of 0.01 mm emerges as the most efficient choice for this setup. However, for coarser meshes, the method fails to converge, suggesting that further investigation is needed. However, this analysis falls beyond the scope of this thesis.

5.3 *Linear Kinematic Hardening*

For this model (see Section 3.4.3), we aim to demonstrate that this multifield implementation also performs well under cyclic loading, unlike the return mapping implementation used as the reference solution. However, since the previous

figures have already extensively proven the accuracy of the multifield approach, we have chosen not to include a reference solution in the following results; both due to its demonstrated reliability and the difficulty of finding a suitable reference for comparison.

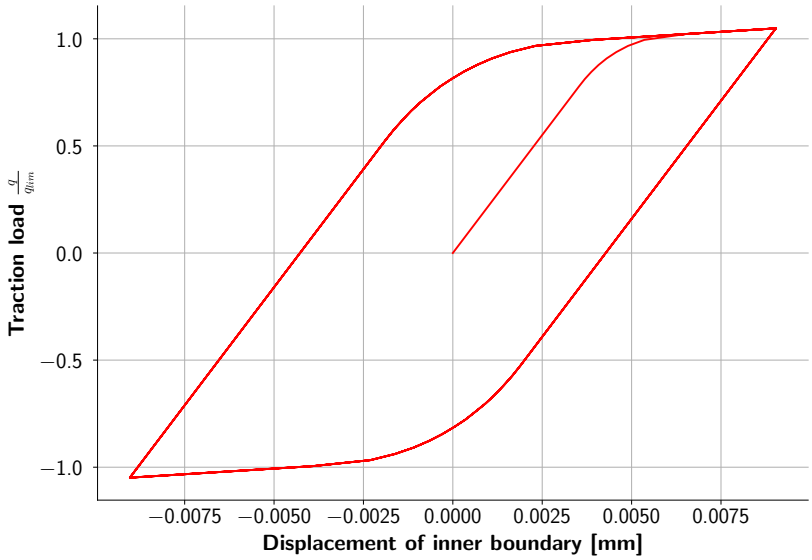

Figure 20: Linear Kinematic Hardening under cyclic loading (5 cycles)

A key observation in the he plot in Fig.20 is that the loading and unloading curves remain consistent across cycles, indicating that the stress required to induce plastic flow does not increase over successive cycles. This behavior aligns with pure kinematic hardening, in which the material retains its initial yield strength but shifts its stress-strain response based on the accumulated plastic deformation. Unlike isotropic hardening, where the yield surface would expand, leading to a progressive increase in the stress required for yielding, this plot shows no significant widening of the loop.

5.4 *Linear Mixed Hardening*

In real-life materials, both isotropic and kinematic hardening typically occur simultaneously under plastic straining. This means that, during plastic deformation, the yield surface both expands and translates in stress space. To capture this more realistic behavior, plasticity models can combine the laws for isotropic and kinematic hardening. This approach is particularly useful for cyclic plasticity applications, where, within a single cycle, kinematic hardening dominates and can account for effects such as the Bauschinger effect. However, over many cycles, the material also exhibits isotropic hardening, causing the peak tension and compression stresses to increase with each cycle until saturation is reached . When a material undergoes both kinematic and isotropic hardening, the yield

surface experiences both translation and expansion. This progressive increase in peak stress and strain in the hysteresis loop due to isotropic hardening is known as cyclic hardening. Cyclic hardening typically occurs over multiple cycles, while kinematic hardening primarily takes place within each cycle [9].

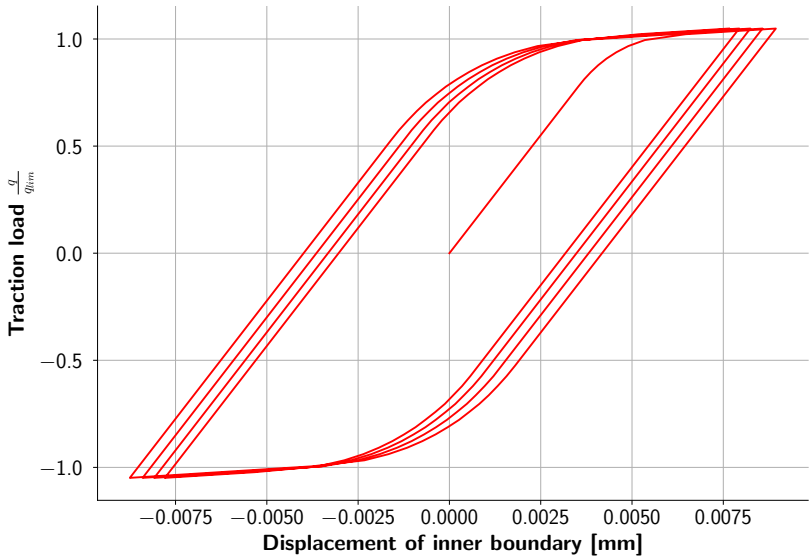

Figure 21: Linear Mixed (Isotropic + Kinematic) Hardening under cyclic loading (5 cycles)

This plot in Fig.21, differently from the previous one, shows clear signs of isotropic hardening. Unlike the previous plot, where the hysteresis loop remained nearly unchanged in size and only translated (indicating kinematic hardening), here we observe a progressive expansion of the loop with each cycle. This behavior suggests that the yield surface is growing as plastic deformation accumulates, requiring increasingly higher stress levels to reach plastic yielding in subsequent cycles.

§ 6 Conclusions

In this thesis, we explored the multifield formulation of small-strain elastoplasticity within the Generalized Standard Materials (GSM) framework, contrasting it with the classical return mapping approach. The multifield formulation presents a thermodynamically consistent and variationally derived alternative that allows for a direct derivation of governing equations from energy potentials, eliminating the need for stress update algorithms traditionally used in computational plasticity.

Through a detailed theoretical development, we demonstrated how the Helmholtz free energy and dissipation potentials fully define the material behavior, enabling an automated and flexible approach to plasticity modeling. Unlike conventional methods that rely on local stress updates and iterative correction schemes, our formulation naturally accommodates higher-order time integration and extends seamlessly to gradient plasticity models, which are not directly compatible with return mapping techniques.

The numerical experiments focused on the multifield approach, validating its accuracy and stability across different problem setups. We examined key aspects such as computational effort, convergence properties, and the impact of mesh refinement on solution quality. The results highlighted that, while the multifield method introduces additional unknowns, it provides a systematic and unified framework that simplifies the implementation of new material models by requiring only the definition of thermodynamic potentials.

Moreover, leveraging modern finite element frameworks such as FEniCSx, we demonstrated how this approach integrates naturally with symbolic computation tools, facilitating the implementation of complex constitutive models. Unlike most existing implementations, which primarily focus on stress updates and tangent operators, our work goes a step further by constructing the formulation directly from energy principles.

Overall, this study underscores the potential of the multifield approach as a powerful alternative to traditional plasticity formulations. By automating the derivation of governing equations through a variational framework, it not only streamlines the modeling process but also opens new possibilities for extending computational plasticity to more advanced material behaviors and complex loading conditions.

Future work could explore the extension of this methodology to finite-strain plasticity, more sophisticated hardening laws, and coupling with damage mechanics, further broadening its applicability in computational mechanics.

Acknowledgments

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Federico Gabrielli

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Appendix

§ A Thermomechanical Framework of Constitutive Relations

In the kinetic theory of gases, the thermodynamic behavior of a macrosystem (such as a mass element of a continuum) is explained through the mechanics of its microsystem (the collection of particles that make up the element). While thermodynamics was primarily concerned with gases, its connection to continuum mechanics was not immediately apparent. However, when applied to other materials, the molar volume must be replaced by strain and the pressure by stress, revealing that thermodynamics and continuum mechanics become inseparable, forming one single branch of science.

In generalized thermodynamics, a key assumption is that the state of a material is fully determined by a specific set of independent variables: the kinematic variables (strain and internal variables) and temperature. The First Law of Thermodynamics asserts that there exists a property (the internal energy) u such that:

$$\dot{u} = \dot{W} + \dot{Q} \tag{A.1}$$

where $\dot{W} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}}$ is the mechanical work input and $\dot{Q} = \nabla \cdot \boldsymbol{q}$ denotes the heat supplied to a volume element. The Second Law of Thermodynamics can be formulated in several ways, but the most practical approach in this context is to state that there exists a property, the entropy s, such that:

$$\dot{s} \geq -\nabla \cdot \left(\frac{\boldsymbol{q}}{\theta}\right) \tag{A.2}$$

where $\frac{q}{\theta}$ is the entropy flux and θ the absolute temperature. Using the product rule for differentiation:

$$\nabla \cdot \left(\frac{\boldsymbol{q}}{\theta}\right) = \frac{1}{\theta}\nabla \cdot \boldsymbol{q} - \frac{1}{\theta^2}\nabla \cdot (\boldsymbol{q}\theta)$$

we can restate the Second Law of Thermodynamics as follows:

$$\theta\dot{s} + \nabla \cdot \boldsymbol{q} - \frac{1}{\theta}\nabla \cdot (\boldsymbol{q}\theta) \geq 0 \tag{A.3}$$

The first two terms $\theta\dot{s} + \nabla \cdot \boldsymbol{q} = \mathcal{D}$ are referred to as the *mechanical dissipation*, while the third term is known as *thermal dissipation*. The thermal dissipation is always non-negative due to the fact that the heat flux is always directed along the negative thermal gradient [13]. For slow processes, the third term becomes negligible in comparison to the first two, leading to the conclusion that the mechanical dissipation must also be non-negative. Therefore, from the First Law of Thermodynamics, it follows that:

$$\dot{u} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \nabla \cdot \boldsymbol{q} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} + \theta\dot{s} - \mathcal{D}$$

$$\implies \quad \mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} + \theta\dot{s} - \dot{u} \geq 0 \tag{A.4}$$

that is the Clausius-Duhem inequality.

Since entropy is a less "measurable" variable, it may be more practical to replace it with the temperature θ, which is more commonly observable, as a state variable. For this reason, it is often preferable to consider the Helmholtz free energy $\psi(\boldsymbol{\varepsilon}, \theta, \boldsymbol{\alpha})$ instead of the internal energy $u(\boldsymbol{\varepsilon}, s, \boldsymbol{\alpha})$ as the thermodynamic potential. The opposite of this potential is defined as the partial Legendre transform (with respect to s) of the internal energy u, given by:

$$-\psi(\boldsymbol{\varepsilon}, \theta, \boldsymbol{\alpha}) = \theta s - u(\boldsymbol{\varepsilon}, s, \boldsymbol{\alpha}) \tag{A.5}$$

Its time derivative is then:

$$-\dot{\psi} = \theta \dot{s} + s\dot{\theta} - \dot{u} \quad \Longrightarrow \quad \theta \dot{s} - \dot{u} = -\dot{\psi} - s\dot{\theta} \tag{A.6}$$

Therefore, by substituting equation (A.6) into the dissipation inequality (A.4), we obtain:

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \dot{\psi} - s\dot{\theta} \geq 0 \tag{A.7}$$

In most plasticity models, isothermal conditions are assumed. This assumption is valid in many practical cases because:

- Plastic deformation often occurs slowly enough for heat generated to dissipate effectively.
- The material is often in thermal contact with its surroundings, which act as a heat reservoir, maintaining constant temperature.

Under this assumption, $s\dot{\theta} = 0$ and we ultimately lead to to equation (1):

$$\mathcal{D} = \boldsymbol{\sigma} : \dot{\boldsymbol{\varepsilon}} - \dot{\psi} \geq 0 \tag{A.8}$$

§ B Perfect Plasticity with Non-smooth Dissipation Potential

In the case of a convex but non-smooth dissipation potential, when formulating the complementary law, it is necessary to introduce the subdifferential

$$\mathcal{A}_I^{dis} \in \partial_{\dot{\boldsymbol{\alpha}}_I} \pi(\dot{\boldsymbol{\alpha}}_I) = \begin{cases} -\sigma_y & \dot{\boldsymbol{\alpha}}_I < 0 \\ [-\sigma_y, \sigma_y] & \dot{\boldsymbol{\alpha}}_I = 0 \\ \sigma_y & \dot{\boldsymbol{\alpha}}_I > 0 \end{cases}$$

It is important to observe that the subdifferential's expression naturally determines the admissible domain for $\mathcal{A}_I^{dis}$, which is the closed and convex set $C = [-\sigma_y, \sigma_y]$.

The dissipation potential for a perfect plasticity model is expressed as

$$\pi^\star(\boldsymbol{\mathcal{A}}_I^{dis}) = \max_{\dot{\boldsymbol{\alpha}}_I} \left\{ \boldsymbol{\mathcal{A}}_I^{dis} : \dot{\boldsymbol{\alpha}}_I - \sigma_y \|\dot{\boldsymbol{\alpha}}_I\| \right\} \tag{B.1}$$

which yields

$$\pi^\star(\boldsymbol{\mathcal{A}}_I^{dis}) = I_C(\boldsymbol{\mathcal{A}}_I^{dis}) = \begin{cases} 0 & \|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| \leq \sigma_y \\ +\infty & \|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| > \sigma_y \end{cases} \tag{B.2}$$

where I_c represents the *indicator function* of the closed and convex set C in the stress space. The dual complementary law, which governs the evolution equation for $\dot{\boldsymbol{\alpha}}$, is then given by:

$$\dot{\boldsymbol{\alpha}}_I \in \partial_{\boldsymbol{\mathcal{A}}_I^{dis}} \pi^\star(\boldsymbol{\mathcal{A}}_I^{dis}) = \partial_{\boldsymbol{\mathcal{A}}_I^{dis}} I_C(\boldsymbol{\mathcal{A}}_I^{dis}) = \begin{cases} 0 & \|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| < \sigma_y \\ \lambda \frac{\boldsymbol{\mathcal{A}}_{I,dev}^{dis}}{\|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\|} & \|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| = \sigma_y \end{cases} \tag{B.3}$$

From this, it is straightforward to establish that the Kuhn-Tucker conditions for maximizing the plastic dissipation closely resemble the normality assumption:

$$\lambda \geq 0, \quad \|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| - \sigma_y \leq 0, \quad \lambda \left(\|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| - \sigma_y \right) = 0 \tag{B.4}$$

Furthermore, it can be shown that the normality hypothesis of plasticity leads to the uniqueness of the solution for the boundary-value problem for an elasto-plastic material [7].

As mentioned earlier in Section 3.3, equation (B.3) represents an alternative formulation of the principle of maximum dissipation, which we will derive in the following, ultimately leading back to equation B.3.

The dissipation potential, with $\boldsymbol{\mathcal{A}}_I^{dis} = \boldsymbol{\mathcal{A}}$ for the sake of notational simplicity, can be expressed as the Legendre transform of its dual, as follows:

$$\pi(\dot{\boldsymbol{\alpha}}_I) = \max_{\boldsymbol{\mathcal{A}}} \left\{ \boldsymbol{\mathcal{A}} : \dot{\boldsymbol{\alpha}}_I - I_D\left(\boldsymbol{\mathcal{A}}\right) \right\} = \max_{\boldsymbol{\mathcal{A}}} \left\{ \mathcal{D} - I_D\left(\boldsymbol{\mathcal{A}}\right) \right\} = \max_{\boldsymbol{\mathcal{A}} \in C} \left\{ \mathcal{D} \right\} \tag{B.5}$$

where the indicator function has been eliminated by introducing the constraint $\boldsymbol{\mathcal{A}} \in C$.

By employing the method of Lagrange multipliers, the maximization in equation (B.5) subjected to the constraint $\boldsymbol{\mathcal{A}} \in C$ (i.e. $\|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| - \sigma_y \leq 0$), can be converted into an unconstrained minimization problem. This is achieved by incorporating the constraint into the objective function, leading to a modified function that is minimized over the unconstrained domain:

$$\mathcal{D}_\lambda \left(\boldsymbol{\mathcal{A}}_I^{dis}, \lambda \right) = -\mathcal{D} \left(\boldsymbol{\mathcal{A}}_I^{dis} \right) + \lambda \left(\|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\| - \sigma_y \right) \tag{B.6}$$

with the Lagrange multiplier $\lambda \geq 0$, the stationarity condition for the modified function reads:

$$\dot{\boldsymbol{\alpha}}_I = \lambda \frac{\boldsymbol{\mathcal{A}}_{I,dev}^{dis}}{\|\boldsymbol{\mathcal{A}}_{I,dev}^{dis}\|} \tag{B.7}$$

which is equivalent to (B.3), subjected to the optimality conditions (B.4). Consequently, the Lagrange multiplier corresponds to the plastic multiplier, linking the dissipation mechanism to the evolution of plastic deformation. This equivalence highlights that the plastic multiplier governs the rate of plastic flow, ensuring consistency between the maximum dissipation principle and the yield criterion.

§ C Return Mapping Formulation

The following return mapping formulation has been derived with guidance from Laura De Lorenzis' Nonlinear FEA lecture notes, which served as a key reference in this process.

Considering a Linear Isotropic Hardening model as outlined in Section 3.4.2, at each quadrature point, we assume that at time t_n the quantities $\boldsymbol{\varepsilon}_n, \boldsymbol{\alpha}_{I,n}, \alpha_{II,n}$. Therefore, we also have:

$$\boldsymbol{\mathcal{A}}_{I,n}^{dis} = -\frac{\partial \psi}{\partial \boldsymbol{\alpha}_{I,n}}(\boldsymbol{\varepsilon}_n, \boldsymbol{\alpha}_{I,n}, \alpha_{II,n}) = \boldsymbol{\sigma}_n = \mathbb{C} : (\boldsymbol{\varepsilon}_n - \boldsymbol{\alpha}_{I,n})$$

Additionally, the incremental displacement $\Delta \boldsymbol{u}_n$(and consequently the incremental strain $\Delta \boldsymbol{\varepsilon}_n$) is given. The objective is to update the state variables at the quadrature point to the new values $\boldsymbol{\varepsilon}_{n+1}, \boldsymbol{\alpha}_{I,n+1}, \alpha_{II,n+1}$, allowing us to compute:

$$\boldsymbol{\mathcal{A}}_{I,n+1}^{dis} = \boldsymbol{\sigma}_{n+1} = \mathbb{C} : (\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\alpha}_{I,n+1})$$

$$\mathbb{D}_{n+1} = \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}}$$

Thus, the local problem reduces to the time integration of the incremental elasto-plastic constitutive law within the given time discretization.

Let's begin by taking the flow rule from Section 3.3 and the hardening law from Section 3.4, and discretizing them in time using the *backward Euler method*:

$$\boldsymbol{\alpha}_{I,n+1} = \boldsymbol{\alpha}_{I,n} + \sqrt{\frac{3}{2}}\Delta\lambda \frac{\boldsymbol{\mathcal{A}}_{I,n+1}^{dev}}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev}\|} \tag{C.1}$$

$$\alpha_{II,n+1} = \alpha_{II,n} + \Delta\lambda \tag{C.2}$$

At time t_{n+1} we can also express the total strain $\boldsymbol{\varepsilon}_{n+1}$ and the dissipative driving forces $\boldsymbol{\mathcal{A}}_{I,n+1}$ as

$$\boldsymbol{\varepsilon}_{n+1} = \boldsymbol{\varepsilon}_n + \Delta\boldsymbol{\varepsilon}_n \tag{C.3}$$

$$\boldsymbol{\mathcal{A}}_{I,n+1} = \mathbb{C} : (\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\alpha}_{I,n+1}) \tag{C.4}$$

where $\Delta\boldsymbol{\varepsilon}_n$ is given (and thus $\boldsymbol{\varepsilon}_{n+1}$ is known). Finally, the time-discrete version of the Kuhn-Tucker conditions at time t_{n+1} reads

$$f_{n+1} = \sqrt{\frac{3}{2}}\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev}\| - (\sigma_{y0} + H\alpha_{II,n+1}) \leq 0, \quad \Delta\lambda \geq 0, \quad \Delta\lambda\, f_{n+1} = 0 \tag{C.5}$$

Let us now define a hypothetical state, known as the trial state, which is obtained by assuming that plastic flow is temporarily halted. This state is characterized by the following equations:

$$\boldsymbol{\alpha}_{I,n+1}^{trial} = \boldsymbol{\alpha}_{I,n}$$

$$\alpha_{II,n+1}^{trial} = \alpha_{II,n}$$

$$\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial} = 2\mu(\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n})$$

$$f_{n+1}^{trial} = \sqrt{\frac{3}{2}} \| \boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial} \| - (\sigma_{y0} + H\alpha_{II,n})$$

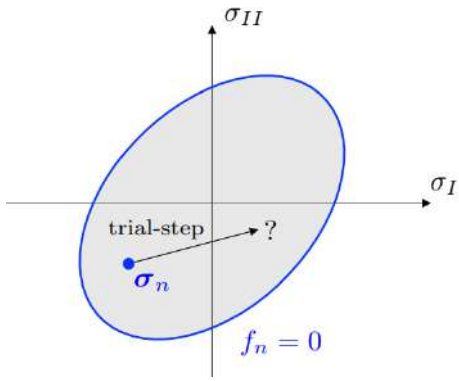

Figure 22: Trial Step, Laura De Lorenzis, Lecture Notes, Nonlinear FEA (2024)

Note that the trial state is solely defined by initial state $\boldsymbol{\varepsilon}_n$, $\boldsymbol{\alpha}_{I,n}$, $\alpha_{II,n}$ and the given incremental strain $\Delta\boldsymbol{\varepsilon}_n$. There are two possible scenarios:

- If $f_{n+1}^{trial} \leq 0$, the trial state is admissible, meaning it satisfies the stress-strain relationship, the flow rule, the hardening law, and the Kuhn-Tucker conditions with $\Delta\lambda = 0$. In this case, the solution is unique, and the trial state represents the solution, indicating that the incremental step is purely *elastic*.

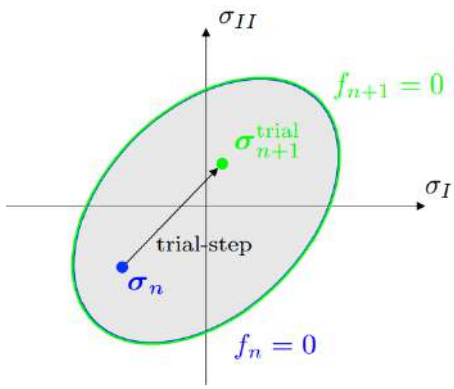

Figure 23: Elastic Step, Laura De Lorenzis, Lecture Notes, Nonlinear FEA (2024)

The quantities of our interest are specifically

$$\boldsymbol{\mathcal{A}}_{I,n+1}^{dis} = \boldsymbol{\sigma}_{n+1} = \boldsymbol{\mathcal{A}}_{I,n+1}^{dis,trial} = \mathbb{C} : (\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\alpha}_{I,n}) \tag{C.6}$$

$$\mathbb{D}_{n+1} = \frac{\partial \boldsymbol{\sigma}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} = \mathbb{C} \tag{C.7}$$

- If $f_{n+1}^{trial} > 0$, the trial state is deemed inadmissible. Consequently, a correction is required, meaning $\Delta\lambda > 0$ to update $\boldsymbol{\alpha}_{I_n+1}$ and $\boldsymbol{\mathcal{A}}_{I,n+1}^{dis}$. Based on the third Kuhn-Tucker condition, it follows that $f_{n+1} = 0$, indicating the incremental step is *plastic*.

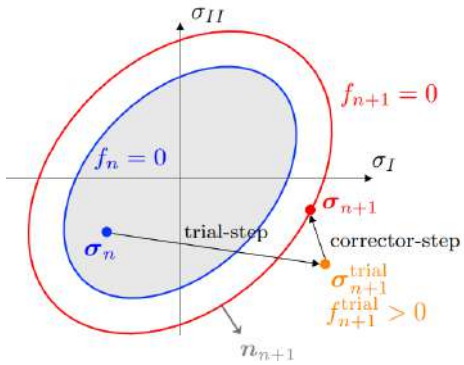

Figure 24: Plastic Step, Laura De Lorenzis, Lecture Notes, Nonlinear FEA (2024)

In Fig.24 we denoted $\boldsymbol{n}_{n+1}$ as

$$\boldsymbol{n}_{n+1} = \frac{\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}$$

Let's see how to correct the trial state in this case. Let us first express $\boldsymbol{\mathcal{A}}_{I,n+1}^{dev}$ in terms of $\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}$ and $\Delta\lambda$ as follows

$$\begin{aligned}
\boldsymbol{\mathcal{A}}_{I,n+1}^{dev} &= 2\mu(\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n+1}) \\
&= 2\mu(\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n}) - 2\mu(\boldsymbol{\varepsilon}_{n+1} - \boldsymbol{\alpha}_{I,n}) \\
&= \boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial} - \sqrt{\frac{3}{2}} 2\mu\Delta\lambda \frac{\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|} \\
&= \boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial} \left(1 - \sqrt{\frac{3}{2}} \frac{2\mu\Delta\lambda}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}\right)
\end{aligned} \tag{C.8}$$

In summary, assuming $\Delta\lambda$ is known, we can determine the true state at time t_{n+1} by correcting the trial state in the following manner:

$$\boldsymbol{\alpha}_{I,n+1} = \boldsymbol{\alpha}_{I,n} + \sqrt{\frac{3}{2}} \Delta\lambda \frac{\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|} \tag{C.9}$$

$$\alpha_{II,n+1} = \alpha_{II,n} + \Delta\lambda \tag{C.10}$$

$$\boldsymbol{\mathcal{A}}_{I,n+1}^{dev} = \boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial} \left(1 - \sqrt{\frac{3}{2}} \frac{2\mu\Delta\lambda}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}\right) \tag{C.11}$$

Finally, $\Delta\lambda$ can be computed by rewriting f_{n+1} as follows

$$\begin{aligned} f_{n+1} &= \sqrt{\frac{3}{2}}\|\boldsymbol{\mathcal{A}}^{dev}_{I,n+1}\| - (\sigma_{y0} + H\alpha_{II,n+1}) \\ &= \sqrt{\frac{3}{2}}\|\boldsymbol{\mathcal{A}}^{dev,trial}_{I,n+1}\| - 3\mu\Delta\lambda - (\sigma_{y0} + H\alpha_{II,n}) - H(\alpha_{II,n+1} - \alpha_{II,n}) \\ &= f^{trial}_{n+1} - (3\mu + H)\Delta\lambda \end{aligned} \tag{C.12}$$

Since from the third of the Kuhn-Tucker conditions it must be $f_{n+1} = 0$, $\Delta\lambda$ reads

$$\Delta\lambda = \frac{f^{trial}_{n+1}}{3\mu + H} > 0 \tag{C.13}$$

Let us now compute the algorithmic tangent modulus by deriving

$$\begin{aligned} \boldsymbol{\mathcal{A}}^{dis}_{I,n+1} = \boldsymbol{\sigma}_{n+1} &= k\,\mathrm{tr}(\boldsymbol{\varepsilon}_{n+1})\boldsymbol{I} + 2\mu(\boldsymbol{\varepsilon}^{dev}_{n+1} - \boldsymbol{\alpha}_{I,n+1}) \\ &= \mathbb{C} : \boldsymbol{\varepsilon}_{n+1} - 2\mu\boldsymbol{\alpha}_{I,n+1} \\ &= \mathbb{C} : \boldsymbol{\varepsilon}_{n+1} - 2\mu\left(\boldsymbol{\alpha}_{I,n} + \sqrt{\frac{3}{2}}\Delta\lambda\boldsymbol{n}_{n+1}\right) \end{aligned} \tag{C.14}$$

This gives

$$\begin{aligned} \mathbb{D}_{n+1} &= \frac{\partial\boldsymbol{\sigma}_{n+1}}{\partial\boldsymbol{\varepsilon}_{n+1}} \\ &= \mathbb{C} - 2\mu\sqrt{\frac{3}{2}}\frac{\partial}{\partial\boldsymbol{\varepsilon}_{n+1}}(\Delta\lambda\boldsymbol{n}_{n+1}) \\ &= \mathbb{C} - 2\mu\sqrt{\frac{3}{2}}\left(\boldsymbol{n}_{n+1}\otimes\frac{\partial\Delta\lambda}{\partial\boldsymbol{\varepsilon}_{n+1}} + \frac{\partial\boldsymbol{n}_{n+1}}{\partial\boldsymbol{\varepsilon}_{n+1}}\right) \end{aligned} \tag{C.15}$$

Thus we need to compute the derivatives of $\Delta\lambda$ and $\boldsymbol{n}_{n+1}$ with respect to $\boldsymbol{\varepsilon}_{n+1}$. Let us start with $\Delta\lambda$. Since

$$\begin{aligned} \Delta\lambda = \frac{f^{trial}_{n+1}}{3\mu + H} &= \frac{\sqrt{\frac{3}{2}}\|\boldsymbol{\mathcal{A}}^{dev,trial}_{I,n+1}\| - (\sigma_{y0} + H\alpha_{II,n})}{3\mu + H} \\ &= \frac{\sqrt{\frac{3}{2}}2\mu\|\boldsymbol{\varepsilon}^{dev}_{n+1} - \boldsymbol{\alpha}_{I,n}\| - (\sigma_{y0} + H\alpha_{II,n})}{3\mu + H} \end{aligned} \tag{C.16}$$

we have

$$\frac{\partial\Delta\lambda}{\partial\boldsymbol{\varepsilon}_{n+1}} = \frac{\partial\Delta\lambda}{\partial\boldsymbol{\varepsilon}^{dev}_{n+1}} : \frac{\partial\boldsymbol{\varepsilon}^{dev}_{n+1}}{\partial\boldsymbol{\varepsilon}_{n+1}} = \frac{\partial\Delta\lambda}{\partial\boldsymbol{\varepsilon}^{dev}_{n+1}} : \mathbb{I}_d \tag{C.17}$$

Here $\mathbb{I}_d$ is the projection tensor on the deviators, i.e.

$$\boldsymbol{\varepsilon}^{dev}_{n+1} = \boldsymbol{\varepsilon}_{n+1} : \mathbb{I}_d \quad \Longrightarrow \quad \frac{\partial\boldsymbol{\varepsilon}^{dev}_{n+1}}{\partial\boldsymbol{\varepsilon}_{n+1}} = \mathbb{I}_d$$

and being $\boldsymbol{n}_{n+1}$ purely deviatoric it is $\boldsymbol{n}_{n+1} : \mathbb{I}_d = \boldsymbol{n}_{n+1}$. So finally we have:

$$\frac{\partial\Delta\lambda}{\partial\boldsymbol{\varepsilon}_{n+1}} = \sqrt{\frac{3}{2}}\frac{2\mu}{3\mu + H}\frac{\boldsymbol{\varepsilon}^{dev}_{n+1} - \boldsymbol{\alpha}_{I,n}}{\|\boldsymbol{\varepsilon}^{dev}_{n+1} - \boldsymbol{\alpha}_{I,n}\|} : \mathbb{I}_d = \sqrt{\frac{3}{2}}\frac{2\mu}{3\mu + H}\boldsymbol{n}_{n+1} \tag{C.18}$$

Next, let us consider $\boldsymbol{n}_{n+1}$, which can be rewritten as

$$\boldsymbol{n}_{n+1} = \frac{\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|} = \frac{\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n}}{\|\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n}\|} \tag{C.19}$$

Recalling the following general result

$$\boldsymbol{n} = \frac{\boldsymbol{x}}{\|\boldsymbol{x}\|} \quad \Longrightarrow \quad \frac{\partial \boldsymbol{n}}{\partial \boldsymbol{x}} = \frac{\|\boldsymbol{x}\| - \frac{\boldsymbol{x}\otimes\boldsymbol{x}}{\|\boldsymbol{x}\|}}{\|\boldsymbol{x}\|^2} = \frac{1}{\|\boldsymbol{x}\|}(\mathbb{I} - \boldsymbol{n}\otimes\boldsymbol{n})$$

we obtain

$$\frac{\partial \boldsymbol{n}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}^{dev}} = \frac{1}{\|\boldsymbol{\varepsilon}_{n+1}^{dev} - \boldsymbol{\alpha}_{I,n}\|}(\mathbb{I} - \boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1}) = \frac{2\mu}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}(\mathbb{I} - \boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1})$$

Thus, using the chain rule,

$$\begin{aligned}
\frac{\partial \boldsymbol{n}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}} &= \frac{\partial \boldsymbol{n}_{n+1}}{\partial \boldsymbol{\varepsilon}_{n+1}^{dev}} : \frac{\partial \boldsymbol{\varepsilon}_{n+1}^{dev}}{\partial \boldsymbol{\varepsilon}_{n+1}} \\
&= \frac{2\mu}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}(\mathbb{I} - \boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1}) : \mathbb{I}_d \\
&= \frac{2\mu}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}(\mathbb{I} - \boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1})
\end{aligned} \tag{C.20}$$

Substituting equations (C.18) and (C.20) into (C.15) we finally obtain

$$\mathbb{D}_{n+1} = \mathbb{C} - \frac{2\mu}{1+\frac{H}{3\mu}}\boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1} - \sqrt{\frac{3}{2}}\Delta\lambda\frac{4\mu^2}{\|\boldsymbol{\mathcal{A}}_{I,n+1}^{dev,trial}\|}(\mathbb{I}_d - \boldsymbol{n}_{n+1}\otimes\boldsymbol{n}_{n+1})$$

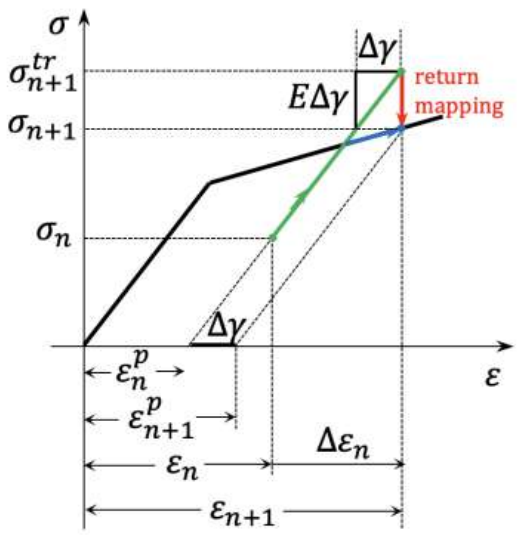

Figure 25: Geometric interpretation of the return mapping algorithm, Laura De Lorenzis, Lecture Notes, Nonlinear FEA (2024)

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